Where Does the Gain Come From? A Controlled Study of Training-Free RoPE Extrapolation to 4K in Diffusion Transformers

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Abstract

Training-free edits of the rotary position embedding (RoPE) let diffusion transformers trained near one megapixel generate 4096×4096 images. Each new method changes several parts of the embedding at once and credits its gain to the newest part, so it is unclear which edit helps. We decompose SEGA, a strong recent method with released code, into its frequency base, a global attention temperature and a content-adaptive per-band allocation. All arms run on FLUX.1-Krea-dev with shared prompts, seeds and sampler; a dev/held-out split was fixed in advance, and every comparison is paired over 187 held-out prompts with bootstrap intervals. SEGA's base and global temperature alone recover all of its ImageReward gain over an NTK base (104%, interval [97, 113]) and tie the released method on three reward models and crop MUSIQ. A tie means the interval covers zero. The allocation adds 2.09 full-frame MUSIQ points alongside reduced excess fine-scale power, without an established causal link. Four controllers derived from efficient coding and contrast constancy do not help: the two evaluated held out tie or lose to released SEGA on every reward and image-quality metric. Code, per-image scores and the scripts that generate every table will be released.

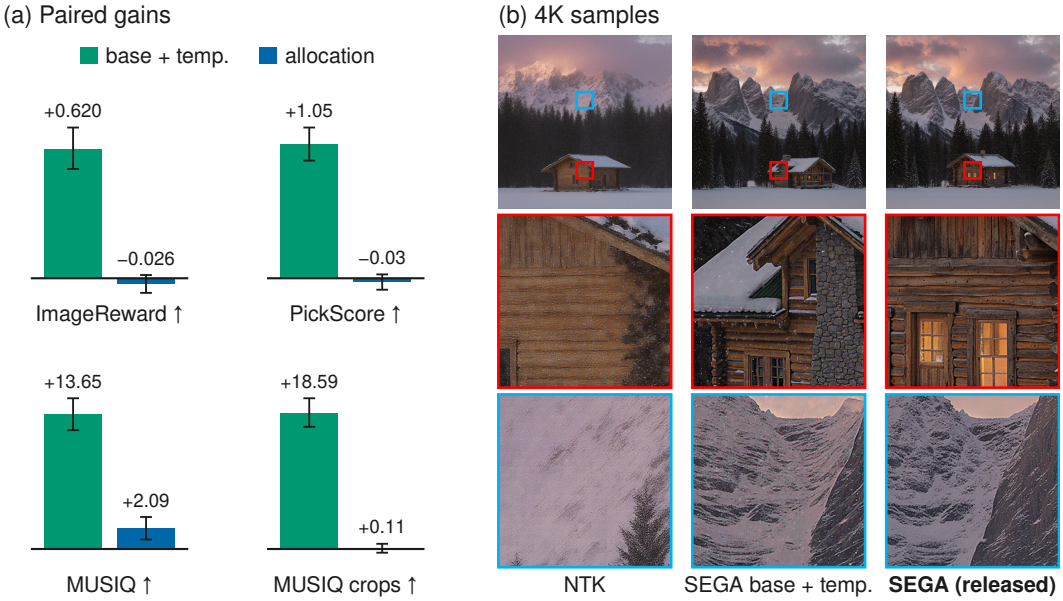


Figure 1: Where SEGA's gain comes from. (a) SEGA base + temp. minus NTK (base + temp.) carries most of the gain; SEGA (released) minus SEGA base + temp. (allocation) adds full-frame MUSIQ. Bars and whiskers are means and nominal 95% paired bootstrap intervals over 187 held-out prompts, seed 0, one scale per metric. (b) Illustrative held-out row 32 was chosen by its cabin caption before viewing; native 384^2 crops (cabin red, mountain cyan) were fixed after viewing at $(2048, 3200)$ and $(2048, 1536)$, identical for every arm.

1 Introduction

Diffusion transformers trained at about one megapixel are now asked for 4096×4096 images, which hold 16 times more tokens than any training image. Training-free methods handle this by editing the rotary position embedding (RoPE) [47] at inference, without new weights or extra sampling passes. They rescale its frequencies [7, 10, 41], schedule the rescaling with the noise level [27] or sharpen attention with a global logit temperature [28, 37]. The most recent methods also scale each rotary band by a gain read from the spectrum of the current latent [45]. Each method is compared as a package with the previous one, and the improvement is credited to its newest component. This paper asks which of these edits actually produces the gain, and whether a per-band law with a principled basis can do better.

Package comparisons cannot answer that question, because each method turns several of the knobs in Table 1 at once. SEGA raises the frequency base, adds a global temperature and adds a content-adaptive per-band allocation in one step, and its paper credits the improvement to the allocation. A comparison against DyPE or an NTK base cannot tell whether the allocation, the temperature or the base did the work. The same confound affects any new per-band law that is tested only against complete methods. Evaluation adds a second difficulty: the reward models used to rank these methods resize their input to 224 pixels, so they cannot resolve the finest octaves of a 4K image. A gain that lives only in those octaves can therefore be invisible to the metrics that decide which method wins.

Table 1: Knobs each arm turns. The shaded row is the reference, SEGA (released); gray marks a context baseline. Table E.1 (Appendix E) expands the table with formulas and sample sizes. Key result: only the per-band column separates SEGA base + temp. from SEGA (released).

Arm	Base	Temp.	Per-band
NTK	NTK	✗	✗
DyPE	ramps in σ	✓	✗
SEGA (released)	SEGA	✓	✓ sensor
SEGA base + temp.	SEGA	✓	✗
Constancy-TM	SEGA	✓	✓ gated
SEGA + TM	SEGA	✓	✓ both

We introduce a controlled decomposition of SEGA [45], a strong recent method with released code, on the checkpoint its authors use (FLUX.1-Krea-dev). The decomposition control, SEGA base + temp., keeps the released frequency base and applies SEGA's reference logit gain $s^{0.16}$ to every query and key dimension, where $s = 4$ is the scale per side. It drops only the per-band allocation, while an NTK base with neither temperature nor allocation serves as the floor. All arms share prompts, seeds, schedule and sampler, and they differ only in the frequencies and gains of the attention logit. We fixed a dev/held-out split before any held-out score was seen, pair every comparison over 187 held-out prompts with bootstrap intervals, and replicate two arms at a second seed. We then ask whether per-band laws with a principled basis improve on SEGA's heuristic. From efficient coding and contrast constancy we derive four controllers, select variants on the dev set under a written gate and evaluate the survivors held out. We do not propose a new state-of-the-art method.

Base and temperature carry most of the gain: of SEGA's gain over the NTK base, they carry 104% of the ImageReward gain (interval [97, 113]), 103% of PickScore and 99% of crop MUSIQ. SEGA base + temp. ties the released method on all three reward models, where a tie means that the paired interval covers zero. The allocation adds +2.09 [+0.96, +3.25] full-frame MUSIQ points alongside less excess power at 8 to 32 pixel wavelengths. This association does not establish a cause, and MUSIQ is a learned quality proxy rather than a sharpness measurement. The principled laws do not help: the two that reached the full held-out set tie or lose to SEGA (released) on every reward and image-quality metric. The closer one, SEGA + TM, loses 0.08 PickScore and 1.13 crop MUSIQ points, and small trajectory diagnostics suggest reasons without identifying causal mediators. We also find that SEGA's printed equations differ from its released code at each of six compared stages, and that the printed law scores worse in a dev pilot. Finally, a lattice failure on FLUX.1-dev appears in a development comparison that also changes guidance, so checkpoint comparisons need matched settings.

To summarize, we decompose a strong training-free 4K method under a fixed held-out protocol and test principled alternatives to its allocation. We highlight our main contributions below.

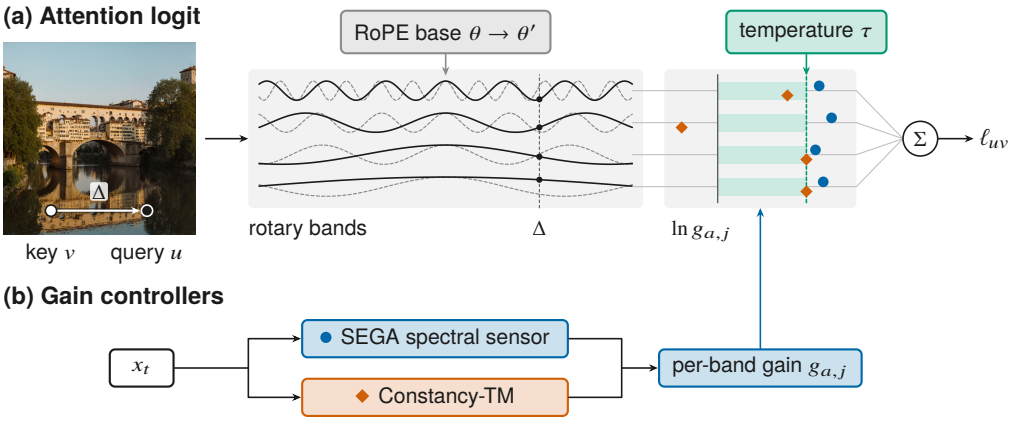


Figure 2: Three knobs act on the logit. (a) Four of the 28 width-axis bands of Equation (1), drawn as unit-amplitude, zero-phase cosines under native base θ (dashed) and released base θ', are read at the key-query offset Δ. Log gains at step 14 of the first held-out prompt, whose SEGA (released) output is shown, are those of SEGA base + temp. (bars), SEGA (released) (circles) and Constancy-TM (diamonds). (b) Either controller reads x_t and sets $g_{a,j}$; Figure E.1 draws both at component level.

- We introduce a base-plus-temperature control and a paired held-out protocol that attribute the gains of training-free RoPE methods at 4K to individual edits.
- We derive four per-band controllers from efficient coding and contrast constancy and find that the two evaluated held out tie or lose to released SEGA on every reward and image-quality metric.
- We show that SEGA's base and global temperature recover 104% [97, 113] of its ImageReward gain over an NTK base, while its allocation adds 2.09 full-frame MUSIQ points.

2 Background

RoPE in FLUX. FLUX [5] packs the autoencoder latent into 2×2 patches, so one token covers 16×16 pixels and a 4096^2 image has a 256^2 token grid (64^2 at 1024^2). Each 128-dimensional attention head has rotary blocks of 16, 56 and 56 dimensions for an index axis and the two spatial axes. Pair j of spatial axis a rotates at $\omega_{a,j} = \theta^{-2j/D}$ with $\theta = 10^4$ and $D = 56$, so wavelengths $2\pi/\omega_{a,j}$ run from about 6 to about 45,000 tokens over the 28 pairs.

The logit as a Fourier series over the offset. Write the query and key coefficients of pair j on axis a as complex numbers $\tilde{q}_{a,j}$ and $\tilde{k}_{a,j}$, and let a method multiply both by a gain $m_{a,j}$ after the rotation, with which it commutes. The logit between a query at position u and a key at position v is then

$$\ell_{uv} = \frac{1}{\sqrt{d_k}} \sum_{a,j} g_{a,j} \left|\tilde{q}_{a,j}\right| \left|\tilde{k}_{a,j}\right| \cos\left(\omega'_{a,j} \Delta^a_{uv} + \phi_{a,j}\right), \qquad g_{a,j} = m^2_{a,j}, \tag{1}$$

where $d_k = 128$ is the head dimension, $\Delta^a_{uv} = p^a_u - p^a_v$ is the offset along axis a, $\omega'_{a,j}$ is the frequency after extrapolation and $\phi_{a,j}$ is the content phase. The logit is thus a Fourier series in the offset, one term per rotary pair, plus a non-rotating index-axis part. Content sets magnitudes and phases; extrapolation changes only the frequencies $\omega'_{a,j}$ and the gains $g_{a,j}$. A gain shared by all pairs, $g \equiv \tau$, is an inverse temperature, and a j-dependent gain reweights the bands of the kernel.

Three knobs. We sort methods by the knobs they turn (Table 1 and Figure 2). *Frequency base*: at 4096^2 the offsets reach $s = 4$ times the native range, so slow pairs see phases never seen in training. Position interpolation divides every frequency by s [10]; NTK-aware scaling raises the base to $\theta' = \theta s^{D/(D-2)}$, which leaves the fastest pair alone and divides the slowest by s [7]; YaRN interpolates only the slow pairs [41]. *Global temperature*: a shared gain $\tau_N = \log N / \log N_{\text{train}}$ offsets the growth of softmax entropy with the key count N [28], and $\tau_N = 1.333$ at 4096^2. *Per-band gain*: a j-dependent $g_{a,j}$ that may vary with the noise level and the image.

The flow schedule at 4K. FLUX samples a rectified flow [34, 35] with the time shift $\sigma' = e^{\mu}/\left(e^{\mu} + 1/\sigma - 1\right)$, where σ is the unshifted noise level and μ grows linearly with the token count [16]. The

Algorithm 1: SEGA (released), one gain update

Input: projected input features h on the $L_h \times L_w$ grid; frequencies ω_a of the released base; scale s.

Output: query/key gains m, one per rotary pair.

```
m_ref = s ** 0.08            # logit gain s^0.16
Z = mean(h, axis=channels)   # sensor, App. E
Z = Z - mean(Z)
spread = 1 - flatness(radial_power(Z)) ** 1.5
m = ones(64)                 # index axis keeps 1
for a in (height, width):
    E = axis_power(Z, a)   # max(L_a//2, 8) bins
    r = rank(log(2*pi / w[a]))   # 0: fastest
    e = interp(log(E), (1 - r) * (K - 1))
    z = (e - mean(e)) / std(e)
    c = tanh(1.5 * z)
    c = c - mean(c)     # Sec. 5.3: 0.15, clamp
    m[a] = max(1, m_ref*(1 - 0.15*spread*c))
return m                # logits scale by m**2
```

Algorithm 2: Constancy-TM, one gain update

Input: latent x_t at noise level σ; previous clean estimate $\hat{x}_0$; native target $P(f, \sigma)$; $\eta = 1$.

Output: query/key gains, one per rotary pair.

```
log_g = full(64, 0.16*log(s))  # SEGA temp.
if x0_hat is None:             # step 0
    return exp(log_g / 2)
for a in (height, width):
    Et = band_energy(x_t, a)  # half octaves
    E0 = band_energy(x0_hat, a)  # App. E.1
    rho = relu(Et / (sigma**2 * C) - 1)
    w = rho / (1 + rho)     # visibility gate
    P = P_native(a, f_b, sigma)  # matched
    A = (f_b >= 1/64) & (f_b <= 1/16)
    c = mean(log(E0[A]) - P[A])  # anchor level
    e = 0.5 * (P + c - log(E0))  # Eq. (3)
    d = eta * w * clip(e, -1, 1)
    # zero for pair wavelengths above L_a
    log_g[a] += interp_logf(d, f_pairs[a])
return exp(log_g / 2)          # q/k gains
```

diffusers default [49] extrapolates that line to $\mu = 11.55$ at 4096[2], so the last evaluated noise level is $\sigma = 0.99974$: every model call sees almost pure noise, and the final Euler step jumps from there to the image. The official DyPE and SEGA pipelines cap μ at 1.15 (last $\sigma = 0.10472$), as do all our arms. This diagnosis is numerical, since we did not measure the default's effect on image quality, but comparisons that mix the two schedules confound position embedding with sampling.

3 Methods under study

All arms share the sampler and differ only in ω' and g of Equation (1). Table 1 summarizes the knobs, and Table E.1 in Appendix E gives formulas, code revisions and sample sizes.

Frequency baselines. NTK is the strengthened NTK base that SEGA builds on, $\theta' = \theta s^{2D/(D-2)}$, which divides the slowest pair's frequency by s^2; with no temperature and no allocation, it is the floor of the decomposition. DyPE [27], ported from the official code, blends position interpolation, NTK and native frequencies with two ramps in σ^2, so fine pairs regain their native frequencies late in denoising. Its cos/sin amplitude $1 + 0.1 \ln s$ acts as a logit gain of 1.30 on the spatial dimensions.

SEGA: paper equations and released code. SEGA [45] sets per-pair gains from its input spectrum around $m_{\text{ref}} = s^{0.08}$, giving more weight to low-energy pairs; Algorithm 1 gives the released update as we ported it. We call this parity-tested released port SEGA (released) and our implementation of the printed equations SEGA (paper eqs.); the two differ at each of six compared stages (Section 5.3).

SEGA base + temp.: the decomposition control. SEGA base + temp. keeps the frequency base of SEGA (released) and applies its reference logit gain $s^{0.16} = 1.248$ uniformly to every dimension. It differs from SEGA (released) in three ways: no per-band shape, no lift in mean gain from the clamp at 1, and a gain of $s^{0.16}$ rather than 1 on the index axis.

3.1 Per-band laws from vision science

Contrast constancy describes nearly stable perceived band contrast above threshold across viewing distances [20]. Efficient coding combines visibility against noise, $S/(S+N)$, with contrast normalization toward natural-image statistics [1, 2, 13, 18, 33, 39, 48] and divisive gain control [9, 23, 46, 51]. We test four controllers inspired by these ideas; Appendix E.1 gives their equations and calibration.

- **Constancy.** A noise-visibility gate weights a bounded log-power correction toward a native-resolution target from the final clean steps, on the NTK base with log-N temperature.
- **Constancy-TM** (Algorithm 2). The same correction uses a target matched to the current noise level and the base and temperature of SEGA (released), with SEGA base + temp. as its control.
- **Alias-T.** A pair's resolvable key count sets its log-N temperature; Alias-T-s matches SEGA's reference geometric mean.

- **SEGA + TM.** The time-matched correction multiplies SEGA's released allocation, $\ln g = \ln g^{\text{SEGA}} + \Delta$. Development selection chose $\eta = 1$ for the held-out run.

4 Experimental setup

Model and sampler. The held-out comparisons use the FLUX.1-Krea-dev transformer [6], the checkpoint loaded by the official DyPE and SEGA scripts, with the FLUX.1-dev text encoders and autoencoder [5]. A 4096^2 image has 65,536 tokens, 16 times the 4,096 of a native 1024^2 image ($s = 4$ per side). Every held-out arm uses 28 Euler flow-matching steps, distilled guidance 4.5, bfloat16, batch size one, the same per-prompt seeds and the capped flow shift $\mu = 1.15$ (Appendix E).

Prompts and split. Of the 195 evaluation captions of Aesthetic-4K [55], rows 0–7 form a development set for variant and hyperparameter choices and rows 8–194 the held-out set ($n = 187$). We fixed this split after an eight-prompt pilot and before inspecting any held-out score; the decomposition hypothesis arose from that pilot and was tested only held out. Seed 0 is used throughout. A seed-1 replication covers rows 8–62 ($n = 55$) for SEGA (released), SEGA base + temp. and Constancy-TM; it and the partial Constancy run use the first rows of the fixed manifest, not random subsamples.

Arms and baselines. Six arms ran on all held-out prompts: NTK, DyPE, SEGA (released), SEGA base + temp., Constancy-TM and SEGA + TM (Table 1). The original Constancy ran on 41 held-out prompts, and SEGA (paper eqs.) and the Alias-T variants only on dev prompts (Appendix D).

Hyperparameter selection. From the second campaign on, a written gate let a variant leave the dev set only with higher ImageReward and full-frame MUSIQ than SEGA (released) on the eight dev prompts and a CLIP drop of at most 0.3, and at most two variants could advance per campaign. Dev numbers are selection evidence only (Appendix D.2).

Metrics. *Alignment and preference*: CLIP score [24, 43] ($100 \max(\cos, 0)$ of ViT-L/14 embeddings), PickScore [32] and ImageReward [53]. These models resize their input to 224 pixels, so we read them as measures of layout and alignment. *No-reference quality*: MUSIQ [30] and CLIP-IQA [50] on the full 4096^2 frame ("full") and averaged over four fixed 1024^2 crops ("crop"). Full-frame MUSIQ (pyiqa 0.1.16) scores all 16,577 patches of the native 32^2 grid and its 224- and 384-pixel views with no cap (Appendix E), so it includes every original pixel but weights context and scale differently from crop MUSIQ. *Spectral fidelity*: RMS and slope errors of luminance power in 11 octave bands from 2^{-12} to 2^{-1} cycles per pixel against the mean profile of the 195 Aesthetic-4K references, and a tiling index (Appendix B). FID and KID [4, 25] are too noisy at our sample sizes to rank methods. No metric measures 4K quality alone, and we ran no human study.

Statistics. Each comparison is a paired difference against SEGA (released) over the same prompts and seed, with a 95% percentile bootstrap interval over 2,000 prompt resamples. Better, worse and tie describe these nominal intervals; a tie means the interval covers zero, not equivalence. We also compute approximate two-sided centered-bootstrap p-values and apply Holm correction separately over the condensed comparisons of Tables 2 and B.1 and over all comparisons of Table B.4; stars mark adjusted $p \leq 0.05$, while intervals and triangles stay nominal. The allocation's full-frame MUSIQ gain survives both families, but its CLIP-IQA gain and SEGA + TM's small CLIP loss do not. These bootstrap tests give approximate error control, and the other tables are descriptive.

Compute. The 1558 retained 4096^2 images took 79 GPU-hours (73 on GH200, 6 on GB200); a held-out image takes a median 188 s on one GH200 GPU and 93 s on one GB200 (Appendix G).

5 Results

5.1 Base and temperature do most of the work

The floor and the control. Table 2 lists every held-out arm. NTK keeps SEGA's family of frequency base and drops both temperature and allocation; it is far worse on every reward and image-quality metric (ImageReward −0.594, full-frame MUSIQ −15.74, crop MUSIQ −18.70). SEGA base + temp. keeps SEGA's released base and global temperature and drops only the allocation. It ties SEGA

Table 2: Held-out results and decomposition. Means over rows 8–194 (seed 0, $n = 187$); bold marks the best mean and underline the second, which alone imply no significant difference. ▲/▼ mark paired 95% intervals against SEGA (released) (shaded) that exclude zero, and a star marks a Holm-adjusted $p \leq 0.05$; DyPE (gray) is context. The bottom block gives paired gains with nominal intervals (base + temp.: SEGA base + temp. − NTK; allocation: SEGA (released) − SEGA base + temp.) and the share of SEGA's gain over NTK carried by base and temperature. Tables B.1 to B.4 expand this table. Key result: SEGA base + temp. ties SEGA (released) on all three reward models and crop MUSIQ, and the allocation adds only full-frame MUSIQ and CLIP-IQA.

Arm	Image-Reward↑	Pick-Score↑	CLIP score↑	MUSIQ full↑	MUSIQ crop↑	CLIP-IQA full↑
Frequency baselines						
NTK	0.596* ▼	22.03* ▼	27.15* ▼	29.49* ▼	39.63* ▼	0.557* ▼
DyPE	1.026* ▼	22.57* ▼	28.76 ▼	43.35 ▼	**60.30*** ▲	**0.632*** ▲
SEGA (released) and its decomposition control						
SEGA (released)	1.190	23.05	29.14	**45.23**	58.33	0.603
SEGA base + temp.	**1.216**	**23.08**	**29.15**	43.14* ▼	58.22	0.594 ▼
Per-band laws (ours)						
Constancy-TM	1.208	23.01	28.87 ▼	42.19* ▼	56.14* ▼	0.578* ▼
SEGA + TM	1.199	22.98* ▼	29.04 ▼	45.02	57.20* ▼	0.606
Decomposition of SEGA (released)'s gain over NTK (paired, $n = 187$)						
Base + temp. gain	+0.620 ▲ [+0.525, +0.724]	+1.05 ▲ [+0.92, +1.18]	+2.01 ▲ [+1.57, +2.48]	+13.65 ▲ [+12.05, +15.28]	+18.59 ▲ [+16.72, +20.63]	+0.037 ▲ [+0.021, +0.053]
Allocation gain	−0.026 [−0.070, +0.016]	−0.03 [−0.09, +0.03]	−0.02 [−0.23, +0.19]	+2.09* ▲ [+0.96, +3.25]	+0.11 [−0.51, +0.71]	+0.009 ▲ [+0.001, +0.018]
Share of base + temp.	104% [97, 113]	103% [97, 109]	101% [90, 112]	87% [80, 94]	99% [96, 103]	80% [59, 98]

(released) on ImageReward (+0.026 [−0.016, +0.070]), PickScore, CLIP and crop MUSIQ, and falls behind on full-frame MUSIQ (−2.09 [−3.25, −0.96]), CLIP-IQA and the spectral error.

Base and temperature carry the reward--model gain. Of SEGA's gain over NTK, base and temperature carry 104% of the ImageReward gain (interval [97, 113]), 103% of PickScore, 101% of CLIP and 99% of crop MUSIQ (Table 2, bottom; Figure 1a). A share above 100% means the allocation's point estimate is negative.

The allocation adds full-frame MUSIQ. The allocation's own contribution, SEGA (released) minus SEGA base + temp., is +2.09 [+0.96, +3.25] on full-frame MUSIQ and +0.009 [+0.001, +0.018] on CLIP-IQA. Its intervals cover zero on all three reward models and on crop MUSIQ. The higher full-frame MUSIQ accompanies lower excess luminance power at 8 to 32 pixel wavelengths (Figure 4). This spectral change is consistent with suppressing fine-scale artefacts, but it does not establish what caused MUSIQ to rise, and neither spectral power nor MUSIQ alone measures perceived sharpness.

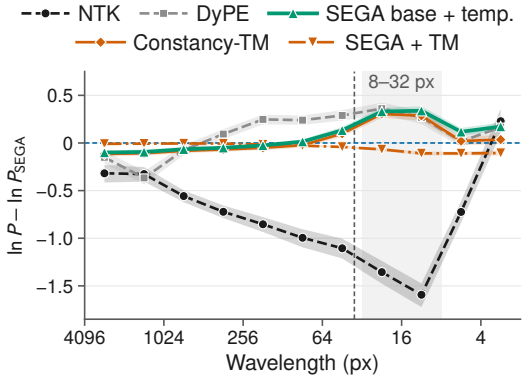


Figure 4: Paired luminance spectra. Removing the allocation (SEGA base + temp.) leaves more octave-band luminance power at 8 to 32 pixel wavelengths (gray band) than SEGA (released), the dashed blue zero line. The band lies above the Nyquist frequency of a 224-pixel view (dashed vertical); intervals are 95% prompt-bootstrap over held-out rows 8–194 ($n = 187$, seed 0). This contrast accompanies the full-frame MUSIQ gap but does not isolate its cause; Figure A.3 in Appendix A adds errors against the reference profile.

DyPE trades reward for crop quality. Against SEGA (released) (Figure 3), DyPE is worse on all three reward models (ImageReward −0.164 [−0.258, −0.087], PickScore −0.49) and better on crop MUSIQ (+1.97 [+1.16, +2.80]) and CLIP-IQA (+0.029). No single metric orders the methods, so we report all of them.

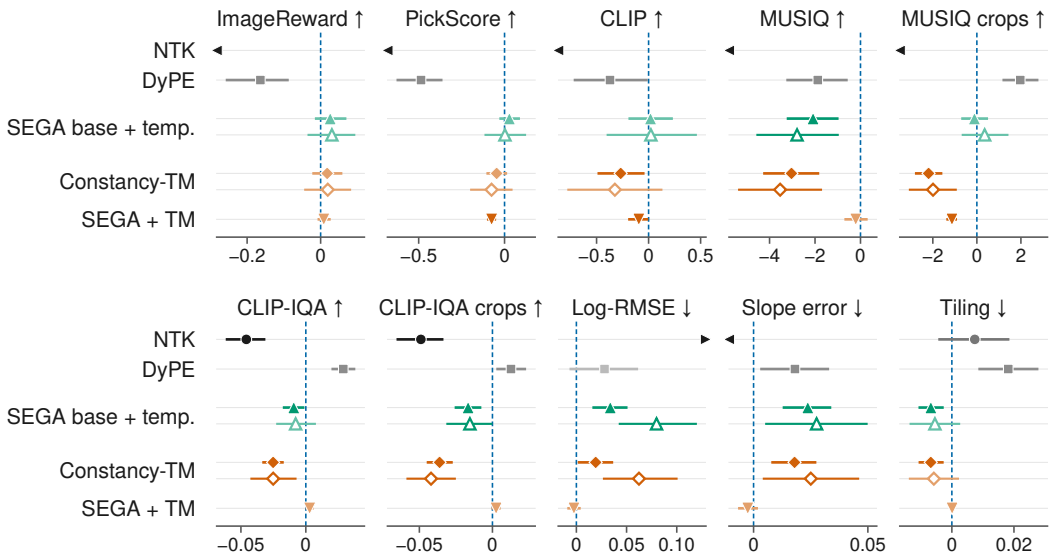


Figure 3: Paired differences from SEGA (released). No per-band law improves on SEGA (released) in any reward or image-quality metric, and DyPE's isolated gains in crop MUSIQ and CLIP-IQA come with losses elsewhere. The dashed blue line is SEGA (released); marks show held-out paired differences with 95% bootstrap intervals (rows 8–194, seed 0, $n = 187$), pale where the interval includes 0, and Table B.1 lists them. Hollow marks: seed-1 replication on rows 8–62 ($n = 55$) for SEGA base + temp. and Constancy-TM (none for SEGA + TM); triangles: NTK beyond the axis limit; arrows: better direction.

5.2 Principled per-band laws do not beat SEGA (released)

Neither held-out law improves on SEGA (released). Neither law that ran on the full held-out set improves on SEGA (released) in any reward or image-quality metric (Table 2, middle block). Constancy-TM ties it on ImageReward and PickScore and is worse on CLIP (−0.27), full-frame MUSIQ (−3.04), crop MUSIQ (−2.19), CLIP-IQA and the spectral error; its one gain is a lower tiling index (−0.0068). SEGA + TM at $\eta = 1$, the only variant to pass the campaign-2 dev gate, ties SEGA (released) on ImageReward (+0.009 [−0.008, +0.027]) and full-frame MUSIQ (−0.21 [−0.71, +0.32]). It loses on PickScore (−0.08 [−0.10, −0.05]) and crop MUSIQ (−1.13 [−1.39, −0.90]), and its CLIP interval, [−0.20, −0.00], ends just below zero. Alias-T and Alias-T-s failed the dev gate on ImageReward (Table D.1).

Against their own controls the corrections cost crop quality. Table 3 adds each allocation to the same base and temperature. Constancy-TM loses to SEGA base + temp. on crop MUSIQ (−2.08 [−2.43, −1.75]) and full-frame MUSIQ (−0.95) while slightly improving both spectral errors (−0.015 and −0.006). Adding the correction on top of SEGA's allocation keeps most of its full-frame MUSIQ gain but still loses crop MUSIQ (−1.02).

Seed 1 repeats most verdicts. On the 55 seed-1 prompts, SEGA base + temp. again ties SEGA (released) on ImageReward, PickScore and CLIP and loses on full-frame MUSIQ (−2.79 [−4.58, −0.95]). Of the 20 nominal verdicts for SEGA base + temp. and Constancy-TM, 17 agree with seed 0 on the same prompts (Table 4).

Diagnostics suggest explanations. Four diagnostics on dev prompts and on logged held-out gains suggest explanations but identify no causal mediator of image quality (Figure F.1, Appendix F). The late-profile visibility gate stays closed through most of sampling; a time-matched target activates the correction earlier, but its better spectral match accompanies lower MUSIQ against SEGA base + temp.. The released allocation is mostly a fixed tilt whose prompt-dependent remainder, which includes each prompt's noise draw, only bounds content dependence; SEGA + TM enlarges that remainder without a held-out quality gain. Three dev probes find weak, prompt-dependent, cross-band responses, and a two-prompt local linear model barely improves on no change; these probes question band-local control without establishing a universal limit on it.

Table 3: Per-band allocation ablation. Paired held-out differences from SEGA base + temp. (n = 187, seed 0) after adding SEGA's allocation (shaded, the released default), the time-matched correction (Constancy-TM) or both (SEGA + TM). Nominal 95% intervals and triangles as in Table 2; no Holm adjustment. Key result: only SEGA's allocation raises full-frame MUSIQ without a crop MUSIQ loss.

Metric	SEGA alloc.	TM corr.	SEGA + TM
ImageReward↑	−0.026 [−0.070, +0.016]	−0.008 [−0.019, +0.003]	−0.017 [−0.060, +0.025]
PickScore↑	−0.03 [−0.09, +0.03]	−0.07 ▾ [−0.10, −0.05]	−0.10 ▾ [−0.18, −0.03]
MUSIQ↑	+2.09 ▴ [+0.96, +3.25]	−0.95 ▾ [−1.67, −0.22]	+1.88 ▴ [+0.80, +2.97]
MUSIQcrop↑	+0.11 [−0.51, +0.71]	−2.08 ▾ [−2.43, −1.75]	−1.02 ▾ [−1.66, −0.42]
Spec. err.↓	−0.034 ▴ [−0.051, −0.016]	−0.015 ▴ [−0.021, −0.008]	−0.036 ▴ [−0.056, −0.015]

Table 4: Seed-1 replication. Paired differences from SEGA (released) on held-out rows 8–62 (n = 55) at seed 0 and seed 1, with nominal triangles. Table C.1 in Appendix C expands the table to ten metrics with intervals; 17 of its 20 verdicts agree. Key result: every verdict shown agrees across seeds, including the full-frame MUSIQ loss of SEGA base + temp..

	SEGA base + temp.		Constancy-TM	
Metric	seed 0	seed 1	seed 0	seed 1
ImageReward↑	+0.070	+0.031	+0.063	+0.019
PickScore↑	+0.05	+0.00	−0.01	−0.08
MUSIQ↑	−3.62 ▾	−2.79 ▾	−4.36 ▾	−3.53 ▾
MUSIQcrop↑	+0.43	+0.35	−1.77 ▾	−1.99 ▾
Spec. err.↓	+0.056 ▾	+0.080 ▾	+0.042 ▾	+0.062 ▾

Native crops show the differences. Figure 5 shows native crops of all six arms on three held-out prompts. The crops are illustrative, because compositions differ between arms.

5.3 SEGA: paper versus code

The printed law is unbounded. With the printed law (Equation (2) and Table 5) the allocation is unbounded. At 4096^2 the physical bin map sends every pair with a wavelength above 256 tokens, 19 of the 28 per axis, to the same frequency bin. The released law scales the spread by 0.15 and clamps the result at 1, and its logit gain stays between 1.00 and 1.61 on our held-out runs.

Table 5: SEGA paper versus code. Our reading of the printed equations against released commit `ddaab08552c7`; Appendix E gives the formulas. Key result: the release shrinks and bounds an allocation that the paper leaves unbounded.

Stage	Printed equations	Released code
Sensor input	latent mean	projected-feature mean
Pair-to-bin map	physical frequency	rank of log period
Spread	ςu, u from tanh z	$0.15\,\varsigma c$, c from tanh $1.5z$
Bounds	none	gains clamped at 1
Index axis	m_{ref}	1
Frequency base	$\theta s^{2D/(D-2)}$, per axis	$\div(1{+}0.1 \ln s)$, longest side

The printed law scores worse in a dev pilot. In the eight-prompt dev pilot (descriptive only; Appendix D.3), our implementation of the printed equations scores worse than the released code on every metric. Freezing the printed allocation into a static tilt destroys images, while freezing the released allocation costs much less (Figure A.4). Readers who reimplement SEGA from the paper will not reproduce the released method, so comparisons against SEGA should state which version they ran.

5.4 Pitfalls in 4K evaluation

224-pixel scorers miss the finest octaves. CLIP, PickScore and ImageReward see a 224-pixel view with Nyquist frequency 0.0273 cycles per original pixel ($2^{-5.2}$). Bands 7 to 10 of our 11 octaves lie entirely above it, and the methods differ in these bands (Figure 4 and Table B.5), which may help explain reward-model ties alongside a full-frame MUSIQ gap. Resizing can keep indirect effects or alias fine patterns, but these scorers cannot directly resolve original detail and do not suffice alone at 4K.

Checkpoint comparisons need matched guidance. Our first campaign used FLUX.1-dev, on which DyPE and Constancy produced a dense halftone lattice over the 4096^2 frame and NTK a featureless blur. Official DyPE code reproduced the lattice on dev prompt 0, seed 42, at guidance 3.5, while the matching FLUX.1-Krea-dev run at guidance 4.5 had no visible lattice (Figure A.2). Our port is therefore not a necessary cause of that failure, but the guidance difference prevents isolating the checkpoint. Like DyPE's main experiments [27], all our held-out comparisons use FLUX.1-Krea-dev; the FLUX.1-dev pilot is descriptive evidence only.

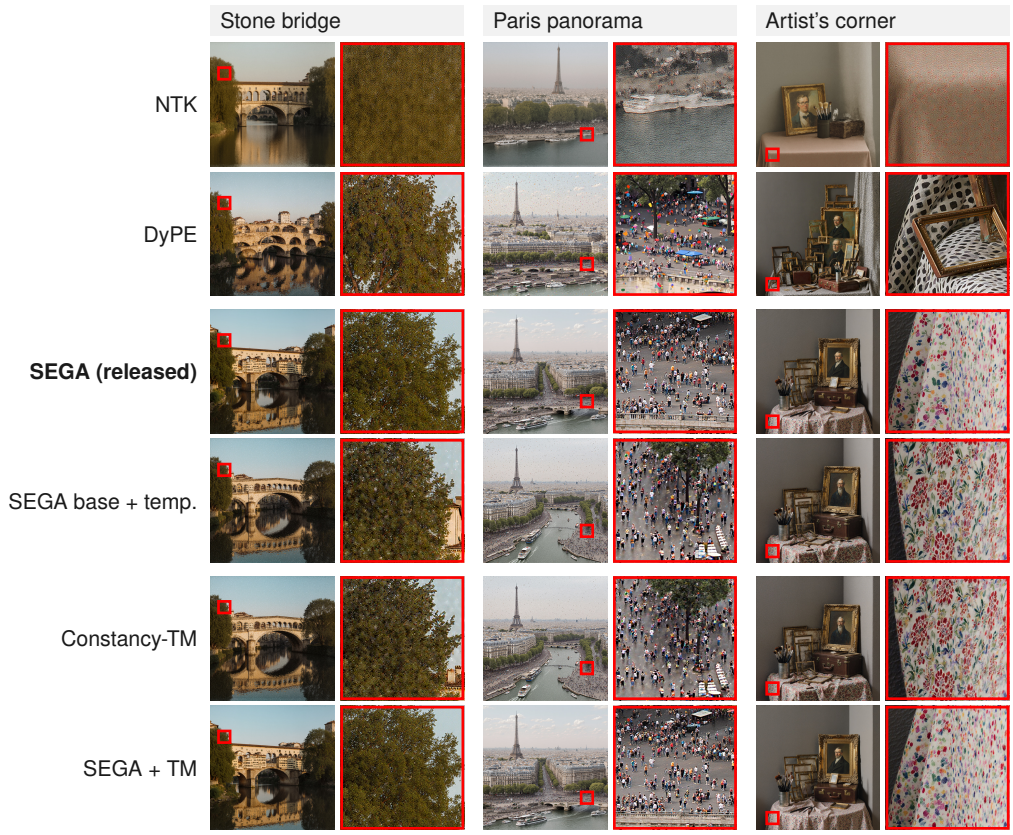


Figure 5: Native crops show what thumbnails hide. Held-out rows 8, 9 and 10 (left to right), the first three held-out prompts, were fixed before any output was viewed; seed 0, each Lanczos-reduced full frame beside its unaltered 384-pixel crop (red box). During review, we chose one shared crop location per row on a fixed grid by maximizing the minimum Laplacian energy across all six arms, each normalized by its 90th percentile over candidate locations. This output-dependent rule favors detail present in every arm and selects no winning method; differing compositions remain a limitation. Full prompts are in Figure A.1.

Attention kernel and GPU. On one GH200 node, forcing FlashAttention-2 makes a sample take 1.54 times as long as cuDNN-first SDPA, while method families on one node differ by at most 0.6% (Table 6); one GB200 GPU is 2.01 times faster than a GH200. Timing comparisons between methods must therefore fix both kernel and GPU. Our records did not log the dispatched kernel or a hash of the loaded weights, so we reconstructed attention policy and transformer identity from code revisions and task scripts, which are not direct runtime records.

Table 6: Kernel and GPU set cost. Median seconds per 4096^2 sample; the FlashAttention-2 timings come from the dev pilot. Table G.1 expands this table. Key result: the kernel changes cost far more than the method does.

Setting	s/image↓	relative
GH200, cuDNN SDPA	188	1.00
GH200, FlashAttention-2	290	1.54
GB200, same code	93	1/2.01

6 Related work

Training-free high-resolution generation. U-Net methods modify convolutional scale [22, 26], fuse patches or scales [14, 42], restructure features [56] or cascade over frequencies [57]. For flow transformers [5, 16, 40], I-Max and HiFlow follow a low-resolution trajectory [8, 15], HRDiT adapts pretrained transformers [54] and Diffusion-4K fine-tunes at 4K [55]; we study embedding edits that add neither training nor sampling passes.

Rotary extrapolation and attention temperature. Interpolation, NTK scaling, YaRN and LongRoPE remap rotary frequencies [7, 10, 12, 41]; frequency analyses separate positional and semantic roles [3, 52]. In diffusion, RIFLEx targets video repetition [58], DyPE schedules remapping [27], UltraImage selects frequencies [59], TIDE controls temperature [36], SEGA adds per-band gains [45]

and AReA adapts locality [44]. Log-key-count temperature is established in variable-size diffusion and Scalable-Softmax [28, 37]; we measure its contribution together with the base.

Spectral views. Coarse-to-fine denoising and harmonic adaptation motivate spectral analyses [11, 17, 21, 29]. Existing interventions target spectra [31], weight training by contrast sensitivity [19] or edit cross-attention Fourier bands [38].

7 Discussion

What to report. A method that changes the rotary base, the temperature and per-band gains at once should report a control that keeps its base and temperature and drops the rest. Here that control leaves a gap of 2.09 full-frame MUSIQ points and no detected reward-model gain, and the allocation's lower excess fine-scale power does not establish a perceptual mechanism. Claims that credit spectral or content adaptivity should be tested against this control with full-resolution metrics (Section 5.4).

Small dev sets mislead. Eight dev prompts were enough to reject clearly worse variants and too few for anything else. SEGA base + temp. looked slightly better than SEGA (released) on dev MUSIQ yet trails by 2.09 points held out, and the variant that passed the second gate lost on PickScore and crop MUSIQ held out. Rerunning SEGA (released) on the same dev prompts moved MUSIQ by more than the margin of the variant that passed the third gate (Appendix D.2). The inspected held-out set cannot serve as fresh confirmation for new variants.

8 Limitations

- **One model and size.** Only FLUX.1-Krea-dev at 4096^2; other shapes, backbones and video are untested.
- **Bundled knobs.** No held-out arm separates the base and temperature changes of SEGA base + temp. relative to NTK.
- **Seeds and prompts.** Most arms have one seed; the seed-1 replication covers two arms on the first 55 held-out prompts, and all prompts come from one caption set.
- **Automatic metrics.** The reward models see 224 pixels, MUSIQ and CLIP-IQA are learned proxies, the spectral error uses a mean reference profile, and we ran no human study.
- **Multiplicity.** Holm adjustment covers the main paired tables only; intervals are nominal, and Table B.4 alone holds 60 comparisons.
- **Small diagnostic samples.** Constancy ran on part of the held-out set, paper-versus-code numbers come from an eight-prompt dev pilot, mechanism probes use at most three dev prompts, and the SEGA CPU study uses another latent space.
- **Confounded checkpoint diagnostic.** Section 5.4 changes both checkpoint and guidance, with official-code evidence on one prompt per checkpoint.
- **Ports.** We port SEGA and DyPE with parity tests against released code, ran official DyPE code only in Section 5.4, and lack the SEGA authors' confirmation of our reading of their paper.

Reproducibility statement

Section 4 fixes prompts, split, seeds, sampler and statistics. Appendix E gives the pipeline, software versions, MUSIQ inputs, pinned SEGA and DyPE commits and every arm's formula (Table E.1), and Appendix E.1 the control laws and calibration. Appendices B and C expand every main table to all ten metrics with intervals and Holm marks; Appendix D lists dev-selection decisions, Appendix F the diagnostic protocols, Appendix G the sampling cost and Appendix H the scripts behind each number. Code, per-image scores and the scripts that generate every table will be released.

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A Additional figures

This appendix expands Figures 4 and 5 and holds the development figures behind Sections 5.3 and 5.4. Figure A.1 shows held-out samples, Figure A.2 the development checkpoint diagnostic, Figure A.3 the octave-band spectra, and Figure A.4 the development paper-versus-code comparison.

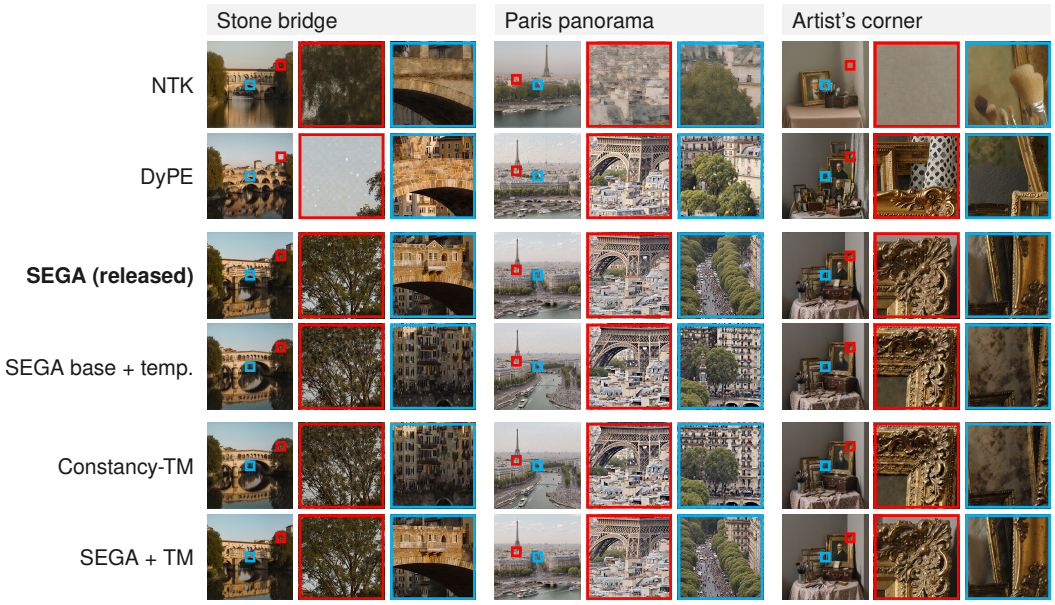


Figure A.1: NTK loses fine detail at 4096^2**.** The other arms render comparable detail, and DyPE shows faint speckle in flat sky on held-out row 8 (one crop checked); compositions differ between arms, so a crop can show different content. Held-out rows 8, 9 and 10 (left to right), the first three held-out rows, were fixed before any output was viewed; seed 0, one sample per arm. Full frames are Lanczos-reduced; native-resolution crops sit at identical coordinates and are never resampled (red: highest-detail tile of the across-arm mean luminance; cyan: image center). Prompts, left to right: "A stone bridge with arched spans reflects golden light on the water, flanked by buildings and a leafy tree under a clear blue sky." "A panoramic view of Paris featuring the Eiffel Tower, the Seine River, and various ornate buildings with greenery and bustling activity, including boats on the water and crowds along walkways." "A cluttered artist's corner features an ornate gold frame, a painting of a man with glasses, several empty picture frames, a vintage suitcase, and various brushes in a tin can, all set against a textured gray wall and draped with patterned fabric."

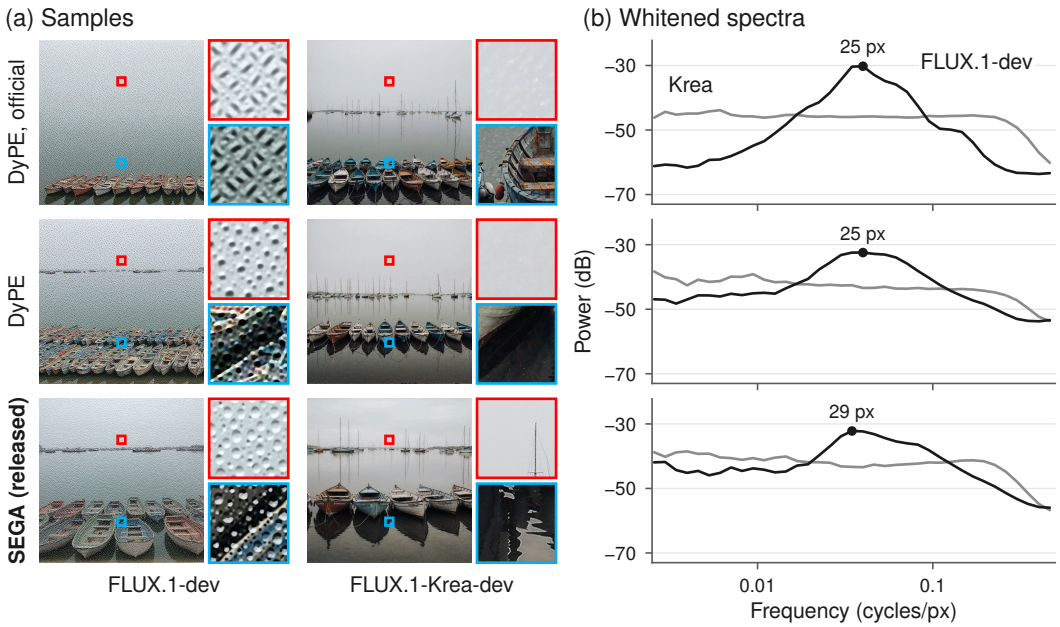


Figure A.2: Lattice and checkpoint check. The development comparison shows a lattice on FLUX.1-dev, but checkpoint and guidance both change. (a) Dev prompt 0 under official DyPE code (seed 42) and our DyPE and SEGA (released) pipelines (seed 0), at guidance 3.5 (FLUX.1-dev) and 4.5 (FLUX.1-Krea-dev). Red and cyan boxes mark native 192-pixel windows at the same place in every image. (b) Whitened radial power spectra of luminance: FLUX.1-dev (black) peaks at the lattice period (dot) and FLUX.1-Krea-dev (gray) does not in these samples. Peak ratios: 36× for the official-code pair; over eight prompts in our pipeline, 12× [7, 20] for DyPE and 5× [2, 11] for SEGA (released) (geometric means, 95% bootstrap intervals). Official-code evidence covers one prompt per checkpoint, and the figure does not isolate a checkpoint effect.

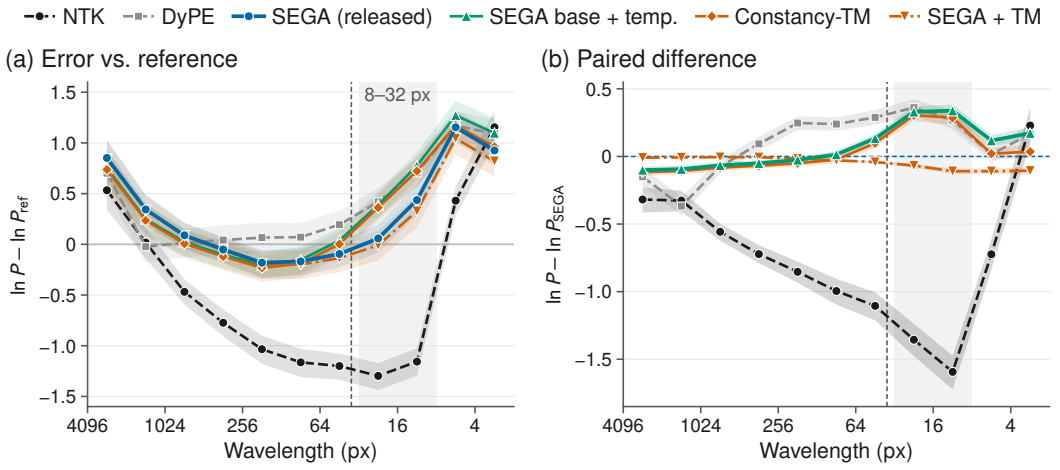


Figure A.3: Spectra against references and SEGA. Without SEGA's per-band allocation, luminance power at 8 to 32 pixel wavelengths (gray band) sits above released SEGA's; Constancy-TM leaves a similar excess, and NTK falls short from 8 to 1024 pixels. (a) Log-power error against the mean ln P of the 195 Aesthetic-4K references, per octave band of linear-sRGB luminance; (b) paired difference from SEGA (released), the dashed blue zero line; positive means more power. Held-out rows 8–194 (n = 187, seed 0) with 95% bootstrap intervals over prompts; the dashed vertical marks the Nyquist frequency of a 224-pixel view.

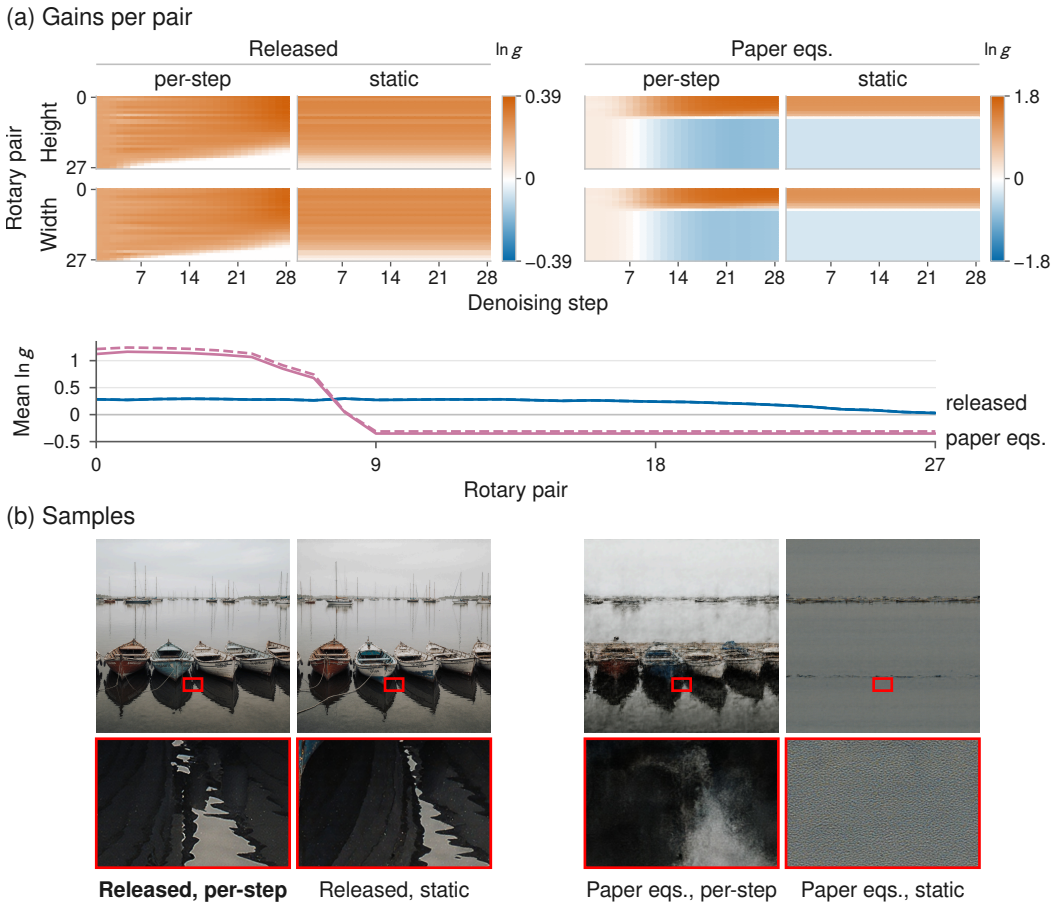


Figure A.4: Printed equations against the release. The printed-law implementation produces larger gain changes than the release and degrades this development pilot. Eight dev prompts (rows 0–7, seed 0), descriptive only. (a) Median log logit gain per rotary pair (pair 0 has the highest frequency) and denoising step. Released and printed-law heatmaps have separate symmetric color scales, so equal colors across the groups do not mean equal magnitudes. The line panel uses one shared axis for mean log gain over prompts, steps and both spatial axes; static variants (dashed) average each method's gains over prompts and steps. (b) Dev prompt 0, seed 0: full frames and a native-resolution crop (red box). **??** gives CLIP score and ImageReward means over the eight dev prompts, with paired differences from SEGA (released) and 95% bootstrap intervals.

B Full held-out results

This appendix expands Table 2: Table B.1 gives its paired differences with intervals, Table B.2 the decomposition with means, Tables B.3 and B.4 all ten metrics and Table B.5 the spectral bands.

Table B.1: Paired held-out differences (expands Table 2). Held-out paired differences against SEGA (released) (rows 8–194, seed 0, $n = 187$); the first row gives the absolute means of SEGA (released). Each cell shows the mean difference and its nominal 95% bootstrap interval; ▲/▼ mark intervals that exclude zero in the better or worse direction; unmarked cells are ties. A star marks a Holm-adjusted $p \leq 0.05$ across this table's paired comparisons, using approximate centered-bootstrap tests. Table B.4 has all ten metrics and a separate Holm family. Key result: base and temperature carry most of SEGA's gain, and no per-band law beats it.

Method	Image-Reward↑	Pick-Score↑	CLIP score↑	MUSIQ full↑	MUSIQ crop↑	CLIP-IQA full↑
SEGA (released)	1.190	23.05	29.14	45.23	58.33	0.603
Decomposition of SEGA (released)						
NTK	−0.594* ▼ [−0.701, −0.494]	−1.02* ▼ [−1.16, −0.89]	−1.99* ▼ [−2.43, −1.55]	−15.74* ▼ [−17.50, −14.00]	−18.70* ▼ [−20.68, −16.86]	−0.046* ▼ [−0.062, −0.031]
SEGA base + temp.	+0.026 [−0.016, +0.070]	+0.03 [−0.03, +0.09]	+0.02 [−0.19, +0.23]	−2.09* ▼ [−3.25, −0.96]	−0.11 [−0.71, +0.51]	−0.009 ▼ [−0.018, −0.001]
Other baseline						
DyPE	−0.164* ▼ [−0.258, −0.087]	−0.49* ▼ [−0.63, −0.36]	−0.37 ▼ [−0.72, −0.01]	−1.88 ▼ [−3.26, −0.56]	+1.97* ▲ [+1.16, +2.80]	+0.029* ▲ [+0.020, +0.038]
Principled per-band laws (ours)						
Constancy-TM	+0.018 [−0.023, +0.059]	−0.05 [−0.11, +0.02]	−0.27 ▼ [−0.49, −0.04]	−3.04* ▼ [−4.28, −1.81]	−2.19* ▼ [−2.81, −1.57]	−0.025* ▼ [−0.034, −0.017]
SEGA + TM	+0.009 [−0.008, +0.027]	−0.08* ▼ [−0.10, −0.05]	−0.09 ▼ [−0.20, −0.00]	−0.21 [−0.71, +0.32]	−1.13* ▼ [−1.39, −0.90]	+0.003 [−0.000, +0.006]

Table B.2 decomposes SEGA's gain over NTK with means and paired gains, Table B.3 gives absolute means and Table B.4 the paired differences for all ten metrics. The spectral rows use the mean octave profile of the 195 Aesthetic-4K reference images as the target; Table B.5 breaks the spectral error into its 11 octave bands. FID and KID are not tabulated: for the seed-1 images (56 prompts per arm, including one dev row), the paired FID difference of SEGA base + temp. against SEGA (released) is +3.26 with interval [−6.86, +13.87], and that of Constancy-TM is +3.74 with interval [−6.72, +14.08].

In band 7, NTK falls below the reference profile (log-power error −1.30); the two arms with SEGA's allocation stay closest to it (SEGA (released) +0.06, SEGA + TM −0.01), and SEGA base + temp. carries excess power (+0.39). These held-out spectral differences lie above the Nyquist limit of the 224-pixel reward-model view (Section 5.4).

Table B.2: Where SEGA's improvement over NTK comes from. Means on the same 187 held-out prompts, paired gains from base and temperature and from the allocation, and the share of the total gain carried by base and temperature, (SEGA base + temp. − NTK)/(SEGA (released) − NTK), with its 95% paired bootstrap interval. A share above 100% means the allocation's point estimate is negative.

	Mean ($n = 187$)			Paired gain		
Metric	NTK	SEGA base + temp.	SEGA (released)	base + temp.	alloc-ation	share of base + temp.
ImageReward↑	0.596	1.216	1.190	+0.620 ▲ [+0.525, +0.724]	−0.026 [−0.070, +0.016]	104% [97, 113]
PickScore↑	22.03	23.08	23.05	+1.05 ▲ [+0.92, +1.18]	−0.03 [−0.09, +0.03]	103% [97, 109]
CLIP↑	27.15	29.15	29.14	+2.01 ▲ [+1.57, +2.48]	−0.02 [−0.23, +0.19]	101% [90, 112]
MUSIQ↑	29.49	43.14	45.23	+13.65 ▲ [+12.05, +15.28]	+2.09 ▲ [+0.96, +3.25]	87% [80, 94]
MUSIQcrop↑	39.63	58.22	58.33	+18.59 ▲ [+16.72, +20.63]	+0.11 [−0.51, +0.71]	99% [96, 103]
CLIP-IQA↑	0.557	0.594	0.603	+0.037 ▲ [+0.021, +0.053]	+0.009 ▲ [+0.001, +0.018]	80% [59, 98]

Table B.3: Held-out means with 95% bootstrap intervals (rows 8–194, seed 0). Constancy is the partial run on 41 prompts (Appendix D).

Metric	SEGA (released) $n = 187$	NTK $n = 187$	SEGA base + temp. $n = 187$	DyPE $n = 187$	Constancy $n = 41$	Constancy-TM $n = 187$	SEGA + TM $n = 187$
ImageReward↑	1.190 [1.087, 1.284]	0.596 [0.456, 0.736]	1.216 [1.118, 1.313]	1.026 [0.910, 1.142]	1.177 [0.953, 1.392]	1.208 [1.110, 1.304]	1.199 [1.096, 1.293]
PickScore↑	23.05 [22.89, 23.21]	22.03 [21.84, 22.21]	23.08 [22.92, 23.24]	22.57 [22.36, 22.75]	22.70 [22.29, 23.08]	23.01 [22.84, 23.16]	22.98 [22.81, 23.13]
CLIP↑	29.14 [28.70, 29.55]	27.15 [26.64, 27.64]	29.15 [28.72, 29.58]	28.76 [28.32, 29.20]	29.46 [28.40, 30.49]	28.87 [28.42, 29.31]	29.04 [28.61, 29.47]
MUSIQ↑	45.23 [43.69, 46.79]	29.49 [28.13, 30.95]	43.14 [41.58, 44.61]	43.35 [41.71, 45.02]	40.55 [37.20, 43.56]	42.19 [40.65, 43.58]	45.02 [43.56, 46.53]
MUSIQcrop↑	58.33 [57.06, 59.63]	39.63 [37.77, 41.33]	58.22 [57.07, 59.44]	60.30 [59.12, 61.53]	60.75 [58.19, 63.01]	56.14 [54.98, 57.36]	57.20 [55.84, 58.52]
CLIP-IQA↑	0.603 [0.590, 0.616]	0.557 [0.546, 0.568]	0.594 [0.581, 0.607]	0.632 [0.618, 0.645]	0.572 [0.544, 0.601]	0.578 [0.566, 0.590]	0.606 [0.593, 0.619]
CLIP-IQAcrop↑	0.586 [0.573, 0.600]	0.537 [0.525, 0.549]	0.569 [0.556, 0.583]	0.599 [0.585, 0.613]	0.543 [0.514, 0.572]	0.550 [0.538, 0.563]	0.588 [0.576, 0.602]
Spec. err.↓	1.068 [1.005, 1.133]	1.285 [1.214, 1.358]	1.102 [1.035, 1.168]	1.096 [1.024, 1.168]	1.337 [1.205, 1.477]	1.088 [1.020, 1.152]	1.066 [1.001, 1.133]
Slope err.↓	0.200 [0.180, 0.221]	0.165 [0.146, 0.185]	0.224 [0.200, 0.247]	0.218 [0.194, 0.242]	0.276 [0.218, 0.338]	0.218 [0.195, 0.240]	0.197 [0.178, 0.218]
Tiling↓	0.0684 [0.0568, 0.0802]	0.0757 [0.0612, 0.0905]	0.0616 [0.0507, 0.0732]	0.0865 [0.0778, 0.0953]	0.0589 [0.0390, 0.0797]	0.0616 [0.0505, 0.0732]	0.0684 [0.0566, 0.0804]

Table B.4: Held-out paired differences against SEGA (released), all metrics. Nominal 95% intervals and direction triangles as in Table B.1; stars mark approximate Holm-adjusted $p \leq 0.05$ across all paired comparisons in this table. Constancy is the partial run on 41 prompts (Appendix D).

Metric	NTK $n = 187$	SEGA base + temp. $n = 187$	DyPE $n = 187$	Constancy $n = 41$	Constancy-TM $n = 187$	SEGA + TM $n = 187$
ImageReward↑	−0.594* ▼ [−0.701, −0.494]	+0.026 [−0.016, +0.070]	−0.164* ▼ [−0.258, −0.087]	+0.018 [−0.089, +0.133]	+0.018 [−0.023, +0.059]	+0.009 [−0.008, +0.027]
PickScore↑	−1.02* ▼ [−1.16, −0.89]	+0.03 [−0.03, +0.09]	−0.49* ▼ [−0.63, −0.36]	−0.07 [−0.28, +0.18]	−0.05 [−0.11, +0.02]	−0.08* ▼ [−0.10, −0.05]
CLIP↑	−1.99* ▼ [−2.43, −1.55]	+0.02 [−0.19, +0.23]	−0.37 ▼ [−0.72, −0.01]	−0.05 [−0.78, +0.68]	−0.27 ▼ [−0.49, −0.04]	−0.09 ▼ [−0.20, −0.00]
MUSIQ↑	−15.74* ▼ [−17.50, −14.00]	−2.09* ▼ [−3.25, −0.96]	−1.88 ▼ [−3.26, −0.56]	−5.09 ▼ [−8.44, −1.74]	−3.04* ▼ [−4.28, −1.81]	−0.21 [−0.71, +0.32]
MUSIQcrop↑	−18.70* ▼ [−20.68, −16.86]	−0.11 [−0.71, +0.51]	+1.97* ▲ [+1.16, +2.80]	+1.91 ▲ [+0.10, +3.80]	−2.19* ▼ [−2.81, −1.57]	−1.13* ▼ [−1.39, −0.90]
CLIP-IQA↑	−0.046* ▼ [−0.062, −0.031]	−0.009 ▼ [−0.018, −0.001]	+0.029* ▲ [+0.020, +0.038]	−0.005 [−0.031, +0.019]	−0.025* ▼ [−0.034, −0.017]	+0.003 [−0.000, +0.006]
CLIP-IQAcrop↑	−0.049* ▼ [−0.066, −0.034]	−0.017* ▼ [−0.026, −0.008]	+0.013 ▲ [+0.003, +0.023]	−0.027 [−0.053, +0.001]	−0.036* ▼ [−0.045, −0.027]	+0.002 [−0.001, +0.006]
Spec. err.↓	+0.216* ▼ [+0.154, +0.287]	+0.034* ▼ [+0.016, +0.051]	+0.028 [−0.007, +0.061]	+0.204* ▼ [+0.127, +0.277]	+0.019 ▼ [+0.001, +0.037]	−0.002 [−0.009, +0.004]
Slope err.↓	−0.035* ▲ [−0.053, −0.018]	+0.024* ▼ [+0.013, +0.034]	+0.018 ▼ [+0.003, +0.033]	+0.061 ▼ [+0.021, +0.101]	+0.018* ▼ [+0.008, +0.027]	−0.003 [−0.007, +0.002]
Tiling↓	+0.0074 [−0.0044, +0.0185]	−0.0068* ▲ [−0.0108, −0.0026]	+0.0182* ▼ [+0.0085, +0.0279]	−0.0099 [−0.0223, +0.0041]	−0.0068* ▲ [−0.0107, −0.0026]	+0.0001 [−0.0007, +0.0008]

Table B.5: Spectral error by octave band. Mean log-power difference from the Aesthetic-4K reference profile on held-out prompts; positive values mean excess power. Bands 7 to 10 lie above the Nyquist frequency of a 224-pixel view of a 4096^2 image. Constancy is the partial run on 41 prompts (Appendix D).

Octave band	0	1	2	3	4	5	6	7	8	9	10
Centre (cyc/px)	$2^{-11.5}$	$2^{-10.5}$	$2^{-9.5}$	$2^{-8.5}$	$2^{-7.5}$	$2^{-6.5}$	$2^{-5.5}$	$2^{-4.5}$	$2^{-3.5}$	$2^{-2.5}$	$2^{-1.5}$
SEGA (released)	+0.85	+0.34	+0.09	−0.05	−0.18	−0.17	−0.10	+0.06	+0.44	+1.15	+0.93
NTK	+0.53	+0.02	−0.47	−0.77	−1.03	−1.16	−1.20	−1.30	−1.16	+0.43	+1.15
SEGA base + temp.	+0.75	+0.25	+0.02	−0.10	−0.21	−0.15	+0.04	+0.39	+0.78	+1.27	+1.10
DyPE	+0.70	−0.02	+0.01	+0.04	+0.07	+0.07	+0.19	+0.42	+0.71	+1.17	+1.09
Constancy	+0.73	+0.34	+0.27	+0.23	+0.21	+0.36	+0.72	+1.37	+1.72	+1.84	+1.64
Constancy-TM	+0.74	+0.24	+0.01	−0.12	−0.23	−0.19	+0.00	+0.36	+0.72	+1.17	+0.96
SEGA + TM	+0.84	+0.34	+0.08	−0.06	−0.20	−0.20	−0.14	−0.01	+0.33	+1.04	+0.82

C Seed-1 replication

This appendix expands Table 4 to all ten metrics with intervals. Table C.1 repeats the comparison for SEGA base + temp. and Constancy-TM on held-out rows 8–62 with a second seed, next to seed 0 on the same 55 prompts. One of the verdicts that changes, crop CLIP-IQA for SEGA base + temp. at seed 1, rests on an interval that excludes zero by less than 10^{-4}, so the agreement count can move by one under another bootstrap seed. Every verdict that changes moves between a tie and a one-sided verdict. Spectral log-RMSE differences grow at seed 1 (SEGA base + temp.: +0.080 against +0.056 at seed 0 on the same prompts; Constancy-TM: +0.062 against +0.042).

Table C.1: Seed 1 repeats most nominal seed-0 verdicts. Paired differences against SEGA (released) on the same 55 held-out prompts (rows 8–62) at seed 0 and at seed 1. 17 of 20 nominal verdicts agree. Intervals and direction triangles use the nominal 95% convention of Table B.1; this descriptive table has no Holm adjustment.

	SEGA base + temp. − SEGA (released)		Constancy-TM − SEGA (released)	
Metric	seed 0	seed 1	seed 0	seed 1
ImageReward↑	+0.070 [−0.011, +0.160]	+0.031 [−0.036, +0.094]	+0.063 [−0.014, +0.142]	+0.019 [−0.045, +0.083]
PickScore↑	+0.05 [−0.07, +0.20]	+0.00 [−0.12, +0.13]	−0.01 [−0.12, +0.12]	−0.08 [−0.20, +0.05]
CLIP↑	−0.02 [−0.54, +0.47]	+0.02 [−0.40, +0.46]	−0.31 [−0.82, +0.17]	−0.32 [−0.78, +0.13]
MUSIQ↑	−3.62 ▾ [−5.84, −1.52]	−2.79 ▾ [−4.58, −0.95]	−4.36 ▾ [−6.57, −2.32]	−3.53 ▾ [−5.38, −1.69]
MUSIQcrop↑	+0.43 [−0.53, +1.40]	+0.35 [−0.69, +1.44]	−1.77 ▾ [−2.82, −0.77]	−1.99 ▾ [−3.09, −0.91]
CLIP-IQA↑	+0.002 [−0.012, +0.017]	−0.008 [−0.023, +0.008]	−0.018 ▾ [−0.034, −0.002]	−0.025 ▾ [−0.043, −0.007]
CLIP-IQAcrop↑	−0.008 [−0.024, +0.009]	−0.015 ▾ [−0.032, −0.000]	−0.034 ▾ [−0.050, −0.015]	−0.042 ▾ [−0.059, −0.025]
Spec. err.↓	+0.056 ▾ [+0.020, +0.091]	+0.080 ▾ [+0.042, +0.120]	+0.042 ▾ [+0.006, +0.077]	+0.062 ▾ [+0.026, +0.101]
Slope err.↓	+0.035 ▾ [+0.018, +0.053]	+0.028 ▾ [+0.005, +0.050]	+0.029 ▾ [+0.013, +0.046]	+0.025 ▾ [+0.004, +0.046]
Tiling↓	−0.0097 ▴ [−0.0157, −0.0038]	−0.0055 [−0.0136, +0.0027]	−0.0098 ▴ [−0.0156, −0.0040]	−0.0058 [−0.0139, +0.0023]

D Partial and dev-only runs (not held-out results)

This appendix expands Sections 4 and 5.3 with the runs behind the dev gate and the pilot. Nothing in it is a held-out result of the main protocol. The first subsection reports a held-out run that covers only part of the held-out set and started before the protocol was fixed; the other two report dev-prompt selection evidence and a descriptive pilot.

D.1 Partial held-out run of Constancy

The original Constancy was generated on 41 of the 187 held-out prompts (seed 0) before the protocol was fixed (Appendix D.2). Every number in this subsection comes from those 41 prompts only, and differences are paired on them. Against SEGA (released) the run ties on the reward models, loses 5.09 full-frame MUSIQ points, gains crop MUSIQ (+1.91) and has a much larger spectral error (+0.204), with excess power in the top octaves and the largest excess of any arm in band 8 (log-power error +1.72; Table B.5). Measured as a share of the gain of SEGA (released) over NTK on the same prompts ($n = 41$), it recovers 110% [100, 120] on crop MUSIQ but 68% [47, 88] on full-frame MUSIQ.

The visibility gate of Section 5.2 opens late on these prompts as on the dev prompts. On the 41 held-out prompts it opens only after the first 20 of 28 steps (median; range 16 to 22), at $\sigma = 0.56$ (range 0.70 to 0.46), and the correction then acts in a median of 8 steps.

This partial run offers suggestive evidence about temperature. NTK and SEGA base + temp. differ in both base and temperature, while Constancy adds to the NTK base the log-N temperature ($\tau = 1.333$) and a per-band correction that stays at zero for the first 20 of 28 steps (median). On its 41 prompts it improves on NTK by +0.688 [+0.479, +0.917] ImageReward and +21.66 [+17.43, +26.06] crop MUSIQ; on the same prompts, the entire SEGA (released) minus NTK gap is +0.670 ImageReward and +19.75 crop MUSIQ. The rows are a nonrandom manifest prefix, the temperature differs from SEGA's, and the correction eventually acts. These observations do not identify the separate base and temperature contributions.

D.2 Dev selection

Table D.1 lists every variant scored on the eight dev prompts, with its mean and its difference from the SEGA (released) reference of the same campaign. These numbers chose what to run held-out; they are not results. Campaign 1 preceded the written gate: it sent SEGA base + temp. to the held-out set as a decomposition control and Constancy-TM as the candidate, and seven chunks of the original Constancy had already run on held-out rows when the protocol was fixed, which is why that arm has only 41 held-out prompts. Campaigns 2 and 3 applied the gate of Section 4. In campaign 3 a fresh SEGA (released) rerun served as the reference. It differs from the pilot run of the same prompts and seed by +0.79 MUSIQ and −0.032 ImageReward, which is more than the MUSIQ margin by which the one passing variant cleared the gate (+0.39); we did not establish the source of the difference. That variant was not run on held-out prompts, and we make no claim about it. The dev set also misjudged the allocation itself: on the eight dev prompts SEGA base + temp. scored +0.24 MUSIQ against SEGA (released), while held out it trails by 2.09.

Table D.1: Dev-set selection (rows 0–7, seed 0, $n = 8$); selection evidence only. Each cell: mean, and below it the difference from the campaign's SEGA (released) reference. [a]Partial held-out run (41 prompts) started before the dev decision. [b]Fresh rerun; the second line is its difference from the pilot run. [c]Passed the dev gate; never evaluated on held-out prompts.

Variant	Image-Reward↑	MUSIQ full↑	CLIP score↑	Pick-Score↑	MUSIQ crop↑	Outcome
Campaign 1 (reference: SEGA (released), pilot run)						
SEGA (released)	1.423 +0.000	42.49 +0.00	30.32 +0.00	23.41 +0.00	58.23 +0.00	reference
NTK + log-N temp.	1.282 −0.141	40.36 −2.13	29.96 −0.36	23.12 −0.29	60.29 +2.05	dev only
Constancy	1.285 −0.138	40.56 −1.93	30.10 −0.22	23.10 −0.31	58.62 +0.39	partial held-out[a]
Constancy-TM on NTK base, $\eta = 1$	1.314 −0.109	41.87 −0.62	30.04 −0.28	23.10 −0.31	57.51 −0.72	dev only
Constancy-TM on NTK base, $\eta = 2$	1.313 −0.110	43.34 +0.85	30.13 −0.19	23.04 −0.37	54.61 −3.63	dev only
SEGA base + temp.	1.444 +0.021	42.72 +0.24	30.32 +0.00	23.26 −0.15	55.93 −2.30	held-out (control)
Constancy-TM	1.432 +0.009	41.65 −0.84	30.24 −0.08	23.25 −0.16	54.57 −3.66	held-out (candidate)
Campaign 2, written gate (reference: SEGA (released), pilot run)						
Alias-T	1.281 −0.142	35.81 −6.68	29.37 −0.95	23.06 −0.35	63.66 +5.43	fails gate
Alias-T-s	1.277 −0.146	44.97 +2.48	30.24 −0.08	23.47 +0.06	58.73 +0.49	fails gate
SEGA + TM, $\eta = 0.5$	1.441 +0.018	42.48 −0.01	30.20 −0.12	23.31 −0.10	58.51 +0.27	fails gate
SEGA + TM, $\eta = 1$	1.449 +0.026	42.51 +0.02	30.34 +0.02	23.33 −0.08	58.28 +0.04	passes, held-out
Campaign 3, same gate (reference: SEGA (released), fresh rerun)						
SEGA (released) (rerun)	1.391 −0.032	43.28 +0.79	30.15 −0.17	23.38 −0.03	58.59 +0.36	reference[b]
SEGA-RMS	1.428 +0.036	43.67 +0.39	30.08 −0.07	23.31 −0.07	58.38 −0.21	passes, not confirmed[c]
SEGA-Uniform	1.374 −0.017	41.71 −1.57	30.73 +0.59	23.29 −0.08	58.07 −0.52	fails gate
SEGA + TM-lag	1.411 +0.019	43.06 −0.22	30.16 +0.01	23.32 −0.05	58.44 −0.15	fails gate

D.3 SEGA paper versus code: pilot numbers

In an eight-prompt pilot on the dev prompts, which is descriptive only, our implementation of the printed equations scores ImageReward 0.964, CLIP 28.14 and full-frame MUSIQ 30.78, against 1.423, 30.32 and 42.49 for the released code (Table D.2). Freezing the printed allocation into a static tilt, a natural ablation of content adaptivity, destroys the images: CLIP falls to 13.06 and logit gains reach 3.49. Freezing the released allocation instead costs much less (CLIP 29.98, ImageReward 1.255). Figure A.4 in Appendix A shows the two gain trajectories and the resulting images. The pilot also ran YaRN, which follows a literal reading of a design note rather than the usual rotation-count convention.

Table D.2: Eight-prompt pilot on the dev prompts (rows 0–7, seed 0), descriptive only. Means; triangles mark nominal paired bootstrap intervals against SEGA (released) that exclude zero in the better or worse direction, which with $n = 8$ is indicative at best. Static variants freeze the allocation instead of reading the current latent.

Method	Image-Reward↑	Pick-Score↑	CLIP score↑	MUSIQ full↑	MUSIQ crop↑	CLIP-IQA full↑	Spectral error↓
SEGA (released)	1.423	23.41	30.32	42.49	58.23	0.557	0.857
SEGA (released, static)	1.255 ▾	23.12	29.98	43.25	57.75	0.559	0.870
SEGA (paper eqs.)	0.964 ▾	22.18 ▾	28.14 ▾	30.78 ▾	42.73 ▾	0.455 ▾	0.944
SEGA (paper eqs., static)	−2.277 ▾	16.93 ▾	13.06 ▾	28.49 ▾	35.53 ▾	0.435 ▾	4.369 ▾
NTK	0.757 ▾	22.37 ▾	27.84 ▾	32.91 ▾	41.41 ▾	0.543	1.220 ▾
YaRN	−0.163 ▾	21.14 ▾	28.06 ▾	42.51	66.69 ▴	0.645 ▴	1.250 ▾
DyPE	1.118 ▾	22.23 ▾	28.59 ▾	44.38	58.32	0.584	0.920
NTK + log-N temp.	1.282 ▾	23.12	29.96	40.36	60.29	0.578	1.038 ▾
Constancy	1.285 ▾	23.10	30.10	40.56	58.62	0.563	1.017 ▾

E Implementation details

This appendix expands Sections 3 and 4: Table E.1 expands Table 1, the paragraphs below give the pipeline, the MUSIQ input, source versions and every arm's formula behind Algorithms 1 and 2, and Appendix E.1 gives the per-band laws.

Table E.1: Arms, their knobs and sample sizes (expands Table 1). Temperature is the logit gain at 4096^2; t/x: varies with the noise level / with the image. Held-out arms ran on rows 8–194, the pilot and dev arms only on the eight dev prompts. The shaded row is the reference, SEGA (released). Key result: SEGA base + temp. matches SEGA (released) in base and temperature and drops only the per-band gain.

Arm	Frequency base	Temperature	Per-band gain	t/x	n
NTK	$\theta\, s^{2D/(D-2)}$	none	none	no / no	187
DyPE	PI/NTK ramps in σ	1.30 (spatial)	none	yes / no	187
SEGA (released)	$\theta\, s^{2D/(D-2)}/(1{+}0.1\ln s)$	$s^{0.16} = 1.248$	sensor, ≥ 1	yes / yes	187
SEGA (paper eqs.)	$\theta\, s^{2D/(D-2)}$	$s^{0.16}$	sensor, unbounded	yes / yes	8 (pilot)
SEGA base + temp.	as SEGA (released)	$s^{0.16}$ on all dims	none	no / no	187
Constancy	as NTK	$\tau_N = 1.333$	gated, late target	yes / yes	41 (partial)
Constancy-TM	as SEGA (released)	$s^{0.16}$	gated, time-matched	yes / yes	187
Alias-T	as SEGA (released)	none	alias count, static	no / no	8 (dev)
SEGA + TM	as SEGA (released)	$s^{0.16}$	SEGA × time-matched	yes / yes	187

Table E.2: Configuration shared by every arm. Gray bands group model, sampling and evaluation settings from the pipeline description below; Table E.1 lists what differs between arms. Key result: the arms differ only in rotary frequencies and gains.

Model	
Transformer	FLUX.1-Krea-dev, 19 double-stream and 38 single-stream blocks
Attention	24 heads of dimension 128; rotary blocks of 16, 56 and 56 dimensions
Text encoders and autoencoder	T5 (512 tokens) and CLIP; FLUX.1-dev autoencoder with tiled decoding
Sampling	
Resolution	4096×4096, a 256^2 token grid
Sampler	28 Euler flow-matching steps, flow shift $\mu = 1.15$, distilled guidance 4.5
Precision and batch	bfloat16, batch size one, per-prompt CPU seed
Evaluation	
Prompts	Aesthetic-4K rows 0–7 (dev) and 8–194 (held-out, $n = 187$)
Seeds	0 for every arm; 1 on rows 8–62 for three arms
Statistics	paired percentile bootstrap, 2,000 resamples, generator seed 0
Software	diffusers 0.39, PyTorch 2.11, pyiqa 0.1.16

Pipeline. The FLUX.1-Krea-dev transformer, loaded through an environment override (`CROPE_TRANSFORMER`), runs inside the FLUX.1-dev pipeline (T5 and CLIP text encoders with 512 T5 tokens, autoencoder with tiled decoding at 4096^2), diffusers 0.39 [49], PyTorch 2.11, bfloat16, batch size one, a per-prompt CPU seed, 28 Euler steps and distilled guidance 4.5. The transformer has 19 double-stream and 38 single-stream blocks with 24 heads of dimension 128, and normalizes q and k before the rotation. Every arm multiplies q and k of every token in every block by one gain vector after the rotation; the vector is shared by blocks and heads and recomputed once per model call. The scheduler patch that holds $\mu = 1.15$ is applied before any configuration is recorded. The per-image

JSON records hard-code the base model name and an attention label; the loaded transformer comes from the task scripts and the attention policy from the code revision (Section 5.4 and Appendix G).

MUSIQ input pipeline. We use pyiqa 0.1.16's KonIQ10k MUSIQ model. The project passes RGB float32 values in $[0, 1]$ without resizing; MUSIQ normalizes to $[-1, 1]$ and combines the native image with aspect-preserving 224- and 384-pixel views, resized by bicubic interpolation with `align_corners=False` and `antialias=False`. Each scale is split into 32×32 patches at stride 32, with SAME padding where needed. The native setting `max_seq_len_from_original_res=-1` retains all patches: a 4096^2 frame contributes 16,384 native patches, plus 49 and 144 resized-view patches, for 16,577 before the class token. A 1024^2 crop contributes 1,217. Neither native square grid needs padding. The spatial embedding's 10×10 hash grid does not downsample the image. We verified these counts by executing the installed preprocessor without loading model weights. Crop MUSIQ averages four separately scored, fixed random 1024^2 crops shared across methods. Full-frame MUSIQ is thus a multiscale quality score that includes every native pixel, not a direct sharpness measurement.

Source versions. Our SEGA comparison uses released commit `ddaab08552c7`. The allocation is in `flux_sega/transformer_flux.py`, lines 103–154; projected-feature sensing follows the input projection in the same file. Our printed-law implementation follows Section 4.1 and equations (3)–(7) of Rajabi et al. [45], with the strengthened base from equation (9) in its Appendix A.2.2. The six-row comparison in Table 5 (Section 5.3) records our interpretation, including how the unspecified index axis and latent input are implemented. We did not obtain confirmation of that interpretation from the authors. Our DyPE port and official-code diagnostic use commit `436fe612fb76`.

Printed SEGA equations. SEGA [45] reads the spectrum of its input at every model call and sets per-pair gains around a reference gain $m_{\text{ref}} = s^{0.08}$. In the printed equations, the energy E_a of the channel-mean latent at pair j's frequency bin is standardized across pairs, $z_{a,j}$, and

$$\begin{aligned} m_{a,j} &= m_{\text{ref}}\big(1 - \varsigma\, u_{a,j}\big), & \varsigma &= 1 - \mathrm{SF}(E_{\text{iso}})^{1.5}, \\ u_{a,j} &= \tanh z_{a,j} - \mathrm{mean}_i \tanh z_{a,i}, \end{aligned} \tag{2}$$

where SF is the flatness of the radial spectrum. Low-energy pairs, usually the fast ones, thus gain attention weight in proportion to how far the spectrum is from flat. Our reading of the printed formulation differs from the release at each of the six stages in Table 5. We call its parity-tested port SEGA (released) and our implementation of the printed equations SEGA (paper eqs.); static variants use the mean allocation over pilot prompts and steps.

DyPE. For spatial axis a with scale s and $\kappa = \sigma^2$, let $b(r) = D \ln\big(64/(2\pi r)\big)/(2 \ln\theta)$ be the pair index that completes r rotations over 64 tokens and $\mu_{r_1,r_2}(j) = 1 - \mathrm{clip}\big((j-\ell)/(h-\ell), 0, 1\big)$ with $\ell = \max(\lfloor b(r_1) \rfloor, 0)$ and $h = \min(\lceil b(r_2) \rceil, D/2)$. Then

$$\begin{aligned} \omega_j^{\text{mix}} &= \tfrac{\omega_j}{s}\big(1 - \mu_{1.25\kappa,0.75\kappa}(j)\big) + \omega_j^{\text{ntk}}\, \mu_{1.25\kappa,0.75\kappa}(j), \\ \omega'_j &= \omega_j^{\text{mix}}\big(1 - \mu_{16\kappa,2\kappa}(j)\big) + \omega_j\, \mu_{16\kappa,2\kappa}(j), \end{aligned}$$

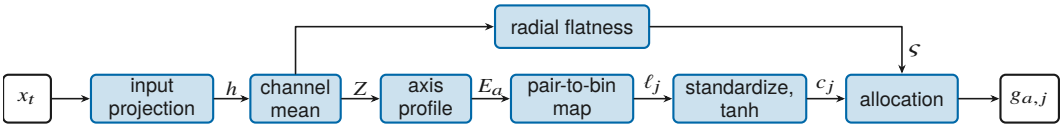
where ω^{ntk} are the NTK-aware frequencies; the spatial cos/sin amplitude is $1 + 0.1 \ln s$, as in the official code at the inspected revision.

SEGA (released). The sensor takes the channel mean Z of the projected input features on the token grid. The axis profile E_a is the one-dimensional FFT power along axis a averaged over the other axis and rebinned into $K_a = \max(\lfloor L_a/2 \rfloor, 8)$ bins. Pair j reads $\ell_j = \ln E_a\big((1-\rho_j)(K_a - 1)\big)$ by linear interpolation, with $\rho_j = j/(D/2-1)$ the normalized rank of its log period; z_j standardizes ℓ_j with the sample standard deviation; and $m_{a,j} = \max\big(1, m_{\text{ref}}[1 - 0.15\,\varsigma\, c_j]\big)$ with $c_j = \tanh(1.5 z_j) - \mathrm{mean}_i \tanh(1.5 z_i)$, $m_{\text{ref}} = \max(s, 1)^{0.08}$ and $\varsigma = 1 - \mathrm{SF}^{1.5}$ from the radially binned two-dimensional power. The index axis keeps gain 1.

E.1 Per-band control laws

Each law sets $\ln g_{a,j} = \ln \tau + \Delta_{a,j}$, a temperature plus a bounded per-band correction. The diffusion noise level supplies the bandwise noise floor.

(a) SEGA spectral sensor

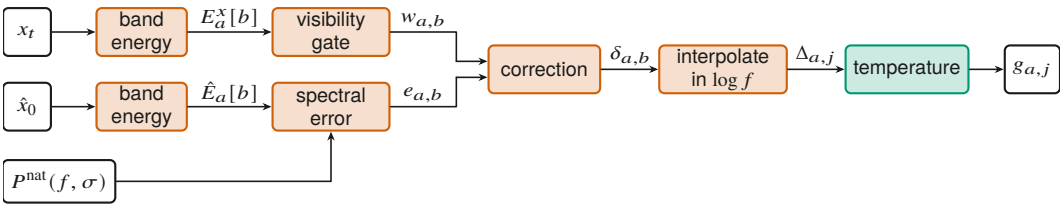


$Z = \mathrm{mean}_c\, h$ $\quad K_a = \max(\lfloor L_a/2 \rfloor, 8)$ $\quad \ell_j = \ln E_a\big((1-\rho_j)(K_a-1)\big)$

$\rho_j = j/(D/2-1)$ $\quad z_j = (\ell_j - \mathrm{mean}_i\, \ell_i)/\mathrm{std}_i\, \ell_i$ $\quad c_j = \tanh(1.5 z_j) - \mathrm{mean}_i \tanh(1.5 z_i)$

$\varsigma = 1 - \mathrm{SF}^{1.5}$ $\quad m_{\mathrm{ref}} = \max(s, 1)^{0.08}$ $\quad m_{a,j} = \max\big(1,\, m_{\mathrm{ref}}[1 - 0.15\,\varsigma\, c_j]\big),\;\; g_{a,j} = m_{a,j}^2$

(b) Constancy-TM controller



$\rho_{a,b} = \max(E_a^x[b]/(\sigma^2 C) - 1, 0)$ $\quad w_{a,b} = \rho_{a,b}/(1+\rho_{a,b})$

$e_{a,b} = \frac{1}{2}\big(P_a^{\mathrm{nat}}(f_b, \sigma) + c_a - \ln \hat{E}_a[b]\big)$ $\quad \delta_{a,b} = \eta\, w_{a,b}\, \mathrm{clip}(e_{a,b}, -1, 1)$

$\ln g_{a,j} = \ln \tau + \Delta_{a,j}$ $\quad \tau = s^{0.16}$

Figure E.1: Both gain controllers at component level. (a) The released SEGA sensor of Figure 2b reads the channel mean of the projected input features, maps the axis profile to rotary pairs by rank and scales its tilt by the radial flatness. (b) Constancy-TM gates a bounded log-power correction toward the time-matched native target and adds it to the SEGA temperature. White boxes are inputs and outputs; the equations repeat Algorithms 1 and 2 and Appendix E.

Constancy. For half-octave bands b, a gate compares the current input's band energy with the known noise floor, $\rho_{a,b} = \max\big(E_a^x[b]/(\sigma^2 C) - 1, 0\big)$ and $w_{a,b} = \rho_{a,b}/(1+\rho_{a,b})$, and an error compares the previous clean estimate's band energy $\hat{E}_a[b]$ with a native-resolution target P^{nat}, the mean log spectrum of the model's own 1024^2 samples on 16 calibration prompts outside the evaluation set:

$$e_{a,b} = \tfrac{1}{2}\big(P_a^{\mathrm{nat}}(f_b) + c_a - \ln \hat{E}_a[b]\big), \qquad \delta_{a,b} = \eta\, w_{a,b}\, \mathrm{clip}(e_{a,b}, -1, 1), \tag{3}$$

where c_a anchors the target level on visible bands of 16 to 64 token wavelengths. The pair correction $\Delta_{a,j}$ interpolates δ in $\log f$, with $\tau = \tau_N$ and $\eta = 1$. A natural spectrum gives $e = 0$ and reduces the law to the global temperature. The target is the native spectrum of the final, nearly clean steps.

Constancy-TM. The same law with a time-matched target: $P^{\mathrm{nat}}(f, \sigma)$ is the clean estimate's native spectrum at the same noise level, tabulated over all 28 native steps, so the anchor no longer needs the gate. Built on the base and temperature of SEGA (released), it isolates the allocation, with SEGA base + temp. as its control.

Alias-T and SEGA + TM. Alias-T applies the log-N temperature pair by pair, taking as a pair's key pool the number of keys it can resolve on the grid (Appendix E); Alias-T-s shifts all its log gains by one constant to match SEGA's reference gain in geometric mean. SEGA + TM multiplies the allocation of SEGA (released) by the time-matched correction, $\ln g = \ln g^{\mathrm{SEGA}} + \Delta$, with $\eta \in \{0.5, 1\}$ on the dev set and $\eta = 1$ held out.

Band energies and calibration. Band energies are densities: with an orthonormal FFT, $P(k) = \sum_c |\mathcal{F}(x_c - \bar{x}_c)(k)|^2$, $E_a(k)$ averages P over the other axis, and $E_a[b]$ averages over half-octave bands b (13 bands at 256 tokens per side). Unit white noise gives $E_a[b] = C = 64$ in expectation, so the noise floor at level σ is $\sigma^2 C$. The gate uses the current input; the error uses the clean estimate of the previous call, $\hat{x}_0 = x_{n-1} - \sigma_{n-1} v_\theta(x_{n-1})$. Corrections are interpolated in $\log f$ between band centers and applied to pairs with frequency above one cycle per grid, which at 4096^2 are pairs 0 to 8 of each axis. Step 0 has no clean estimate and uses the temperature alone, and $|\ln g - \ln \tau| \le \eta$ always. The native targets come from 1024^2 samples of the same model on 16 Pick-a-Pic prompts [32] disjoint

from the evaluation set: the late target averages the three steps with $\sigma < 0.3$, and the time-matched target keeps all 28 steps and is interpolated in σ.

Alias-T. Alias-T applies the log-N temperature of Section 2 pair by pair. A pair of wavelength $\lambda_{a,j}$ tokens under the released base, on a grid of L tokens per side ($L = 64$ at native size), gets the logit gain

$$g_{a,j} = \frac{\ln\big(n(\lambda_{a,j}, L)\, L\big)}{\ln\big(n(\lambda_{a,j}, 64) \cdot 64\big)}, \qquad n(\lambda, L) = \min\big(L, \max(1, L/\lambda)\big),$$

where $n(\lambda, L)$, the pair's alias count, is the number of its wavelengths across the grid, bounded by 1 and L; the index axis takes τ_N. The gains are static and content-free, with no correction and no further temperature. Alias-T-s shifts every $\ln g$ by one constant so that the mean over all 128 dimensions equals SEGA's reference log gain, $0.16 \ln s$.

F Mechanism diagnostics

This appendix expands the diagnostic paragraph of Section 5.2 with the figure, numbers and protocols behind it.

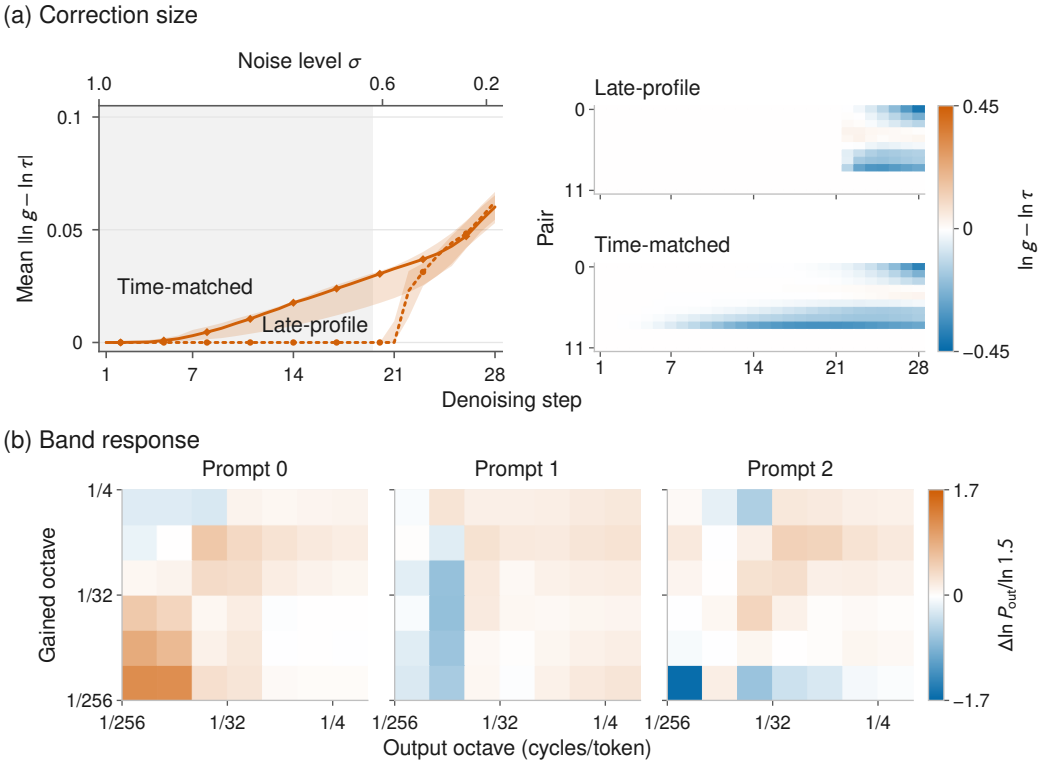


Figure F.1: Dev-prompt mechanism diagnostics. The late-profile correction starts late, and the response to a band-gain change differs across prompts. (a) Mean $|\ln g - \ln \tau|$ over spatial rotary pairs per step: median and interquartile band over eight dev prompts, seed 0, with both targets on the NTK base. In the gray region the late-profile correction is exactly zero in all eight prompts (steps 1 to 19, $\sigma \geq 0.64$). *Right:* median corrections for pairs 0–11; pairs 12–27 are identically zero. (b) Width-axis response matrices for dev prompts 0 to 2, seed 0. Each cell is the same-seed finite-difference change in final-latent log power divided by $\ln(1.5)$ when one input octave receives a 1.5× logit gain on both spatial axes throughout sampling. The matrices share one zero-centerd color scale and are not averaged across prompts; no seed-to-seed uncertainty is estimated.

The following gate and transfer-probe measurements use development prompts; the time-matched and released-allocation summaries use logged held-out trajectories. These diagnostics do not identify causal mediators of image quality.

The visibility gate opens too late. Constancy's correction is exactly zero until a band with a wavelength of 16 to 64 tokens holds more than twice the noise power ($w > \frac{1}{2}$). On the eight dev prompts it first becomes nonzero at $\sigma = 0.60$ and is nonzero on all of them only by $\sigma = 0.46$ (Figure F.1a). For most of sampling, including the steps that fix the layout, Constancy is thus the NTK base with a log-N temperature: the gate is principled, but at 4096^2 the bands it guards stay below the noise floor until late.

A native-resolution target improves spectral match while lowering MUSIQ. The time-matched target opens the gate early. On the held-out Constancy-TM runs, corrections exceed 0.1 in log gain after the first 8 steps (median), and 70% of all nonzero corrections are negative: they lower the mid pairs 6 to 8 (-0.24 at the last step) and, late, the finest pairs 0 to 2 (-0.22). These corrections reduce the gains of the corresponding rotary bands. Against SEGA base + temp., output spectral errors shrink slightly while both MUSIQ scores fall. Matching a native spectrum and maximizing a learned quality score are different objectives; the measurements do not show which individual gain changes cause the score losses.

SEGA's allocation is mostly a fixed tilt. Across the held-out prompts the released logit gain stays between 1.00 and 1.61. A fixed tilt over pairs explains 56% of its variance and a tilt plus a per-step profile 80%. The prompt-dependent remainder, 20% overall and 7.3% from step 14 on, is an upper bound on content dependence, because it also contains each prompt's noise draw. At the last step the finest pair has a logit gain of 1.47 and the 6 slowest pairs per axis sit at the clamp of 1. A law that adapts more to content therefore has little to copy: SEGA + TM raises the prompt-dependent share to 37% and gains nothing held out. A CPU study of the printed equations finds the same structure on another latent space (Appendix F).

The response to gain changes is weak and prompt-dependent. Every law assumes that raising a band's gain raises its output power. A probe on three dev prompts that scales one octave's gain by 1.5 (Figure F.1b) finds a clear same-octave width-axis response for one prompt and a near-zero one for the other two, off-diagonal responses spread across octaves, and a weak height-axis response. On two dev prompts, a local linear model predicts half-step band-power changes little better than no change (Appendix F). These probes question a simple band-local response model, but their small sample does not establish a universal limit on per-band control.

CPU study of the printed SEGA law. Before running any model we measured the printed allocation (Equation (2)) on forward-noised latents of the FLUX.2 autoencoder for 20 DIV2K crops. A fixed tilt over pairs explained 71% of the variance of the modulation, tilt plus timestep 91%, leaving at most 9.2% for content; the flatness gate ς was explained to $R^2 = 0.83$ by the timestep alone and to 0.78 by a generic $1/f$ field. On our held-out FLUX.1-Krea-dev runs the released gate ς is likewise explained to $R^2 = 0.89$ by the timestep. The channel mean that the printed sensor reads keeps 1.7% of the latent power. In a blur sweep at $\sigma = 0.15$ over 10 test crops, the printed law's contrast between fine and coarse pairs changed by 12% between no blur and a blur of 2 tokens, while an oracle version of the constancy law changed 2.4-fold. These measurements motivated the constancy design; they use the printed law, another autoencoder and forward-noised rather than sampled latents, so we use them only as context for Section 5.2.

Probe protocol. The transfer probe of Figure F.1 reruns a dev prompt with the gain of one input octave multiplied by 1.5 on both spatial axes, with the same seed, and reports the finite-difference elasticity $\partial \log P_{\text{out}} / \partial \log g_{\text{in}}$ for six input and seven output octaves of the final latent and image. It used the NTK base with the log-N temperature, three dev prompts and one seed. The local-response experiment applied half-step perturbations at steps 6, 14 and 22 of two dev prompts and compared the observed change in band power with the prediction of a locally fitted linear response: the median error relative to predicting no change was 0.94 (range 0.71 to 1.35; 9 of 12 cases below 1). It tested a single octave, below the resolution of the token grid, so it does not cover all rotary bands. On the width axis of FLUX.1-Krea-dev, the mean diagonal elasticity of the transfer probe is 0.47 for one prompt but -0.03 and -0.04 for the other two, against a mean off-diagonal response of 0.09.

Confounded cross-checkpoint probe. Raising the fastest octave's gain gave a width-axis elasticity of 0.11 on FLUX.1-Krea-dev and -3.64 on FLUX.1-dev for one dev prompt. Guidance and code

revision also differed, so we draw no conclusion about checkpoint-dependent response from the sign change.

Allocation controls on the dev set. A third dev campaign replaced the allocation of SEGA (released) by a uniform gain with the same root-mean-square value, or kept its shape and removed the rise in mean gain caused by the clamp. On the eight dev prompts the uniform control changed MUSIQ by −1.57 and the shape-only control by +0.39, while a rerun of SEGA (released) itself moved MUSIQ by +0.79. These differences are within run-to-run variation and were not taken to held-out prompts.

Diagnostics not reported. The project also measured run-to-run repeatability of the gain histories, the sensitivity of SEGA's sensor to channel-mean shifts at artificial states, attention-operator approximations and native-resolution semantic consistency. None bears on the claims of this paper, and we do not report them.

G Compute and timing

This appendix expands Table 6 with the cost of every retained image.

On the same GH200 node and method, forcing FlashAttention-2 makes a 4096^2 sample take 1.54 times as long as cuDNN-first SDPA (medians 290 s and 188 s), while method families on one node differ by at most 0.6%; one GB200 GPU is 2.01 times faster than a GH200 (Table G.1). The forced-FlashAttention-2 comparison uses development-pilot runs and is descriptive. Timings of different methods are comparable only with the kernel and the GPU fixed.

Table G.1 gives the sampling cost of the 1558 retained 4096^2 images, 79 GPU-hours in total (73 on GH200, 6 on GB200), with a median peak allocation of 40.8 GB. The second campaign's dev runs and the held-out SEGA + TM run used one GB200 node; every other run used GH200 nodes, and every forced-FlashAttention-2 timing at 4096^2 comes from the eight-prompt pilot. The GPU of each image was recovered from task and fleet logs, and its attention policy from the code revision pinned by its implementation hash, because the records carry a constant attention label (Section 5.4). The SDPA column of Table G.1 gives the requested backend policy (cuDNN first: cuDNN, then FlashAttention, then memory-efficient); the dispatched kernel was not logged. The SEGA family comprises released and paper-equation SEGA with their static ablations; the SEGA + TM family, SEGA + TM and its dev-only allocation controls; the Constancy family, Constancy and its dev sweeps, Constancy-TM, the temperature-only ablations, Alias-T and Alias-T-s. Node c619-102 ran 14% slower than the other GH200 nodes under both kernels (median power 665 W versus 836 W while generating), so it is listed separately; the forced-FA2 YaRN and DyPE pilots ran only there. The native rows run FLUX.1-Krea-dev without RoPE changes; the first image of each worker process carries a one-off warm-up (4.1 s with cuDNN first, 0.9 s with FA2), which widens the cuDNN-first IQR at 1024^2. The official DyPE script uses the default SDPA dispatch of diffusers. Timings exclude model loading (89 s per worker process) and PNG writing, except the single official DyPE run.

Table G.1: Sampling cost is set by the attention kernel and the GPU, not by the method. Seconds per image cover prompt encoding, 28 denoising steps and VAE decoding; IQR is the interquartile range; peak is `torch.cuda.max_memory_allocated`; GPU-h sums the per-image seconds; *n* counts images. SDPA policies, method families and the separately listed node are defined in the text.

SDPA	Methods	*n*	Nodes	Median (s)	IQR (s)	Peak (GB)	GPU-h
4096^2, *GH200 (10 nodes)*							
FA2, forced	NTK	8	1	289.8	289.8–289.9	40.8	0.6
FA2, forced	SEGA family	23	3	289.8	289.7–290.1	40.8	1.9
FA2, forced	Constancy family	16	2	290.4	289.9–290.5	40.8	1.3
cuDNN first	NTK	160	6	187.2	186.2–187.5	40.8	8.3
cuDNN first	DyPE	167	6	187.9	187.6–188.1	40.8	8.7
cuDNN first	SEGA family	197	9	187.8	187.5–188.2	40.8	10.3
cuDNN first	SEGA + TM family	24	2	189.4	184.9–189.6	40.8	1.3
cuDNN first	Constancy family	532	8	187.9	187.6–188.6	40.8	27.8
4096^2, *GH200 node c619-102 (listed separately)*							
FA2, forced	YaRN, DyPE	16	1	329.6	329.5–329.7	40.8	1.5
cuDNN first	all methods	195	1	214.6	214.0–215.3	40.8	11.6
4096^2, *GB200 (1 node, 4 GPUs)*							
cuDNN first	SEGA + TM family	204	1	93.3	92.5–94.3	40.8	5.3
cuDNN first	Constancy family	16	1	93.2	92.1–93.3	40.8	0.4
native 1024^2, *GH200, FLUX.1-Krea-dev*							
FA2, forced	no RoPE change	16	1	5.47	5.47–5.48		
cuDNN first	no RoPE change	8	2	4.94	4.93–5.93	36.3	
4096^2, *GH200, official DyPE script*							
default	DyPE (official code)	1	1	193.3			

H Reproducibility checklist

This appendix expands the reproducibility statement of the main text.

- **Numbers.** `paper/tables/make_tables.py` recomputes every held-out paired difference from the archived per-prompt scores, checks it against the registered analysis output (largest absolute difference below 10^{-15}), and writes every table and every number cited in the text. `compute_table.py` and `mechanism_numbers.py` do the same for timings and mechanism statistics. Scratch inputs are snapshotted next to the scripts.
- **Protocol.** Dev rows 0–7 and held-out rows 8–194 of the Aesthetic-4K evaluation captions; seed 0; seed-1 replication on rows 8–62; selection gate and its outcomes in Appendix D.2.
- **Generation.** Every image has a JSON sidecar with the full configuration, code revision, implementation hash, executed noise levels, per-step gains, wall time and peak memory.
- **Statistics.** Paired percentile bootstrap, 2,000 resamples, generator seed 0, with nominal per-metric intervals. Stars in Tables B.1 and B.4 mark approximate centered-bootstrap tests adjusted by Holm within each table; the other descriptive tables retain nominal intervals.
- **Third-party code.** SEGA and DyPE are re-implemented from their released code at pinned revisions with numerical parity tests; DyPE's official code was also run end to end for the checkpoint diagnosis.
- **Not done.** No human study; one model family; one resolution; one seed for most arms; FID and KID too noisy at these sample sizes to rank methods.